

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397262
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Ludlow; Daryl J.

Devices, systems, and methods for electrochemically purifying hydrogen

Abstract

Hydrogen gas purifier electrochemical cells, systems for purifying hydrogen gas, and methods for purifying hydrogen gas are provided. The cells, systems, and methods employ double membrane electrode (DMEA) electrochemical cells that enhance purification while avoiding the complexity and cost of conventional cells. The purity of the hydrogen gas produced by the cells, systems, and methods can be enhanced by removing at least some intermediate gas impurities from the cells. The purity of the hydrogen gas produced by the cells, systems, and methods can also be enhanced by introducing hydrogen gas to the cells to replenish any lost hydrogen. Water electrolyzing electrochemical cells and methods of electrolyzing water to produce hydrogen gas are also disclosed.

Inventors:	Ludlow; Daryl J. (Lake George, NY)
Applicant:	Ludlow; Daryl J. (Lake George, NY)
Family ID:	1000008781203
Appl. No.:	17/934341
Filed:	September 22, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240100475 A1	Mar. 28, 2024

Publication Classification

Int. Cl.: B01D53/32 (20060101); C25B1/04 (20210101); C25B15/08 (20060101)

U.S. Cl.:

CPC **B01D53/326** (20130101); **C25B15/085** (20210101); B01D2256/16 (20130101);
C25B1/04 (20130101)

Field of Classification Search

CPC: B01D (53/326); C25B (1/02)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3489670	12/1969	Maget	N/A	N/A
9478822	12/2015	Blanchet	N/A	N/A
10722841	12/2019	Ukai et al.	N/A	N/A
2004/0211679	12/2003	Wong et al.	N/A	N/A
2007/0246373	12/2006	Ludlow et al.	N/A	N/A
2010/0243475	12/2009	Eisman et al.	N/A	N/A
2017/0233878	12/2016	Yamakura et al.	N/A	N/A
2019/0336911	12/2018	Ludlow et al.	N/A	N/A
2021/0151725	12/2020	Madden et al.	N/A	N/A
2021/0285427	12/2020	Nakaue et al.	N/A	N/A
2022/0136117	12/2021	Molter	205/440	C01B 3/50

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
4006209	12/2021	EP	N/A
WO-2022172120	12/2021	WO	B01D 53/326

OTHER PUBLICATIONS

Sedlak, et al., “Hydrogen Recovery and Purification Using the Solid Polymer Electrolyte Electrolysis Cell,” International Journal of Hydrogen Energy, vol. 6, pp. 45-51, 1980. cited by applicant

Venugopalan, et al.. “Electrochemical Pumping for Challenging Hydrogen Separations,” ACS Energy Letters 2022 7 (4), 1322-1329. cited by applicant

Fishel, et al., “Electrochemical Hydrogen Pumping,” High Temperature Polymer Electrolyte Fuel Cells, Chapter 24, Li, et al. (eds.), Springer, 2016, pp. 527-540. cited by applicant

International Search Report from corresponding PCT application PCT/US2023/072803, dated Jun. 13, 2024, 5 pages. cited by applicant

Written Opinion from corresponding PCT application PCT/US2023/072803, dated Jun. 13, 2024, 9 pages. cited by applicant

Primary Examiner: Chung; Hosung

Attorney, Agent or Firm: Tech Valley Patent, LLC

Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

(1) This invention is generally related to the electrochemical purification and/or compression of hydrogen gas. Specifically, aspects of the invention include electrochemical cells, systems, and methods of purifying and/or compressing hydrogen gas employing one or more membrane electrode assemblies (MEAs) in a single MEA cell and avoiding external handling of gas flows between separate MEA cells.

Description of Related Art

(2) As known in the art, there are many uses for high-purity hydrogen, that is, hydrogen gas having a hydrogen content greater than 99.99 percent, by volume. However, hydrogen gas is commonly found mixed with other, undesirable gases, for example, nitrogen, argon, carbon dioxide, oxygen, and carbon monoxide, among others. Thus, there is a need in the art for improved methods of isolating the hydrogen gas from the undesirable gases to provide a purer form of hydrogen gas.

(3) Hydrogen purification is not easily accomplished. Hydrogen gas is generally difficult to separate from other gases due to, among other things, the hydrogen molecule being relatively small and hydrogen gas being flammable. Existing means for purifying hydrogen gas from undesirable non-hydrogen gases include molecular sieves, membranes, palladium membranes, and electrochemical hydrogen pumps (EHPs).

(4) Molecular sieves separate hydrogen molecules by selective adsorption, preferentially retaining some molecules more favorably than others. In many cases, however, molecular sieve adsorption systems have undesirably insignificant impact on hydrogen gas purity. One such case is the separation of nitrogen gas (N₂) and hydrogen gas (H₂).

(5) Palladium selectively allows only hydrogen atoms to pass through the metal, resulting in the evolution of highly pure hydrogen. However, palladium is expensive, the process requires compressed gas, and high hydrogen recovery rates require high pressure and/or large masses of palladium.

(6) Electrochemical hydrogen pumps (EHPs) selectively extract hydrogen from mixtures of hydrogen gas and other gases, such as, nitrogen and argon. However, typically, other undesirable gases diffuse across the pump's membrane and result in limited hydrogen gas purity. As a result, one might use multiple individual electrochemical hydrogen pumps, each EHP having associated housings, cell stacks, feed conduits, and exhaust conduits, among other separate hardware and control systems, in series to attempt to purify a hydrogen gas stream twice, and thus increase the hydrogen gas purity. However, such configurations require multiple separate electrochemical pump cell stacks and multiple sets of electrochemical stack hardware connected together. This undesirably results in increased system complexity and increased costs.

(7) Another approach for purifying hydrogen is to increase membrane thickness in the electrochemical cell. However, increasing membrane thickness typically is limited to only decrease impurity diffusion across the membrane as a function of the thickness of the membrane. Other approaches for purifying hydrogen gas, require undesirable gas compression, undesirable multiple pumps, recover less hydrogen gas, consume more energy, and/or do not produce the high purity hydrogen gas required by today's hydrogen gas users, such as, the semiconductor industry.

(8) Accordingly, there is a need in the art for improved hydrogen purification systems, methods, and devices.

SUMMARY OF THE INVENTION

(9) The embodiments of the present invention, in their various aspects, address this recognized need by providing improved hydrogen purification that meets, and can exceed, the hydrogen gas purities required by various applications. Aspects of the present invention employ a unique combination of membrane electrode assemblies (MEAs) or "double" MEAs (DMEAs) that have shown to provide the enhance hydrogen gas purities that today's users require.

(10) One embodiment of the invention is a hydrogen gas purifier cell comprising or including: a first membrane electrode assembly (MEA) comprising: a first anode positioned to contact a first gas stream having a first hydrogen gas content and a first impurity gas content, the first anode containing a catalyst, for example, a platinum group-containing catalyst, adapted to oxidize at least some of first hydrogen gas content to produce hydrogen ions and electrons; a first electrolyte, for example, an acidic electrolyte, positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the first anode; and a first cathode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the first cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content greater than the first hydrogen gas content and a second impurity gas content less than the first impurity gas content; and a second MEA comprising or including a second anode positioned to receive the second gas stream from the first cathode of the first MEA, the second anode containing a catalyst adapted to oxidize at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; a second electrolyte, for example, an acidic electrolyte, positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the second anode; and a second cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte of the second MEA, the second cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions and electrons to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content.

(11) In one aspect, the purifier cell may further comprise at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second gas stream. For example, the at least one passage may be located between the first cathode and the second anode. In one aspect, the at least one passage located between the first cathode and the second anode may comprise spaces or voids between mating surfaces of the first cathode and the second anode. In another aspect, the purifier cell may further include a gas-permeable layer or gas diffusion layer (GDL) between the first cathode and the second anode, and the GDL may provide the at least one passage for discharging at least some of the second gas stream. In another aspect, a gas-distribution or flow field insert, with or without a GDL, may be positioned between a first cathode and a second anode to promote or enhance the distribution of the second gas stream across the surface of the second anode. The flow field insert may be an electrically-conductive porous or a perforated plate, for example, a porous or perforated metallic plate, or a screen-like insert, for example, a metallic screen-like insert, positioned and adapted to provide at least some gas distribution about the surface of the second anode. In another aspect, the at least one passage for discharging at least some of the second gas stream may be at least one channel in close proximity to of the first cathode, the second anode, or both.

(12) In another aspect, the purifier cell may further comprise or include at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream. For example, the at least one passage may be located between the first cathode and the second anode. In one aspect, the at least one passage may be spaces or voids between mating surfaces of the first cathode and the second anode. In one aspect, the purifier cell may further include a gas-permeable layer or gas diffusion layer (GDL) between the first cathode and the second anode, and the GDL may provide the at least one passage for introducing hydrogen-containing gas to the second gas stream. In another aspect, a gas-distribution or flow field insert, with or without a GDL, may be positioned between the first cathode and the second anode to promote or enhance the distribution of the second gas stream across the surface of second anode. The flow field insert may be a porous or perforated plate, for example, a porous or perforated metallic plate, or a screen-like insert, for example, a metallic screen-like insert, positioned and adapted to provide at least some gas distribution about the surface of the second anode.

(13) In another aspect, the at least one passage for introducing hydrogen-containing gas to the

second gas stream may comprise at least one channel in close proximity to the first cathode, the second anode, or both.

(14) In one aspect, the first gas stream may have a first gas pressure and the third gas stream may have a third gas pressure, wherein the third gas pressure is greater than the first gas pressure. In other aspects, the third gas pressure may be less than the first gas pressure.

(15) Another embodiment of the invention is a hydrogen gas purifying system comprising or including: at least one hydrogen gas purifier cell as disclosed herein; and at least two electrically conductive plates, one of the at least two plates mounted to a first end of the at least one hydrogen gas purifier cell, and one of the at least two plates mounted to a second end of the at least one hydrogen gas purifier cell, opposite the first end. For example, in one aspect, the at least one hydrogen gas purifier cell may comprise a plurality of hydrogen gas purifier cells, for example, a stack of hydrogen gas purifier cells.

(16) Another embodiment of the invention is a method for reducing an impurity gas content of a gas stream having a hydrogen gas content and an impurity gas content, the method comprising or including: introducing a first gas stream having a first hydrogen content and a first impurity gas content to a first anode containing a catalyst; in the first anode, catalytically oxidizing at least some of the first hydrogen gas content to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions and at least some of the impurity gas content through a first electrolyte to a first cathode containing a catalyst; in the first cathode, catalytically reducing the at least some of the hydrogen ions transferred through the first electrolyte to produce a second gas stream having a second hydrogen content greater than the first hydrogen content and a second impurity gas content less than the first impurity gas content; introducing the second gas stream to a second anode having a catalyst; in the second anode, catalytically oxidizing at least some of the second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions produced at the second anode and at least some of the second impurity gas content through a second electrolyte to a second cathode; and in the second cathode, catalytically reducing the at least some of the hydrogen ions transferred through the second electrolyte to produce a third gas stream having a third hydrogen content greater than the first hydrogen content and a third impurity gas content less than the first impurity gas content.

(17) In one aspect, the method may further include removing at least some of the second gas stream to produce a modified gas stream having a non-hydrogen gas partial pressure lower than a partial pressure of the non-hydrogen gas in the second gas stream. In another aspect, introducing the second gas stream to the second anode comprises introducing the modified gas stream to the second anode. In one aspect, removing at least some of the second gas stream may be practiced by removing at least some of the second gas stream through a passage between the first electrolyte and the second electrolyte, for example, the passage may be located between the first cathode and the second anode. In one aspect, the passage for removing at least some of the second gas stream may be spaces or voids between mating surfaces of the first cathode and the second anode. In another aspect, removing the at least some of the second gas stream may be practiced by removing the at least some of the second gas stream through a gas-diffusion layer (GDL) and/or flow field insert positioned between the first cathode and the second anode. In another aspect, removing the at least some of the second gas stream may be practiced by removing the at least some of the second gas stream through at least one channel in close proximity to the first cathode, the second cathode, or both.

(18) In another aspect, the method may further include introducing some hydrogen gas to the second gas stream, for example, a “make-up” gas stream. In one aspect, introducing some hydrogen gas to the second gas stream may replenish at least some hydrogen gas removed from the second gas stream. In one aspect, the make-up hydrogen gas stream may comprise at least some of the third gas stream having a third hydrogen content. For example, the third gas stream may be introduced to the second gas stream by diffusion through the second electrolyte. This diffusion

through the second electrolyte may be referred to as “back diffusion” of at least some of the third gas stream having the third hydrogen content through the second electrolyte to the second gas stream.

(19) In one aspect, by employing the electrochemical cells and methods disclosed herein, the purified hydrogen gas produced, for example, the third impurity gas content, may be at least 100 times lower, by volume, than the impurity content of the input gas content, for example, the first impurity gas content. In another aspect, the impurity gas content of the hydrogen gas produced may be at least 1,000 times lower, 10,000 times lower, 100,000 times lower, or even 1,00,000 times lower or less, than the impurity content of the first hydrogen gas stream.

(20) In one aspect, the impurity gas content of the hydrogen gas produced by any of the methods, cells, and systems of the present invention, for example, in the third gas stream, may be at most 100 parts per million [ppm], that is, the third gas stream may contain at most 100 ppm of impurity gas. In other aspects of the invention, the impurity gas content of the hydrogen gas produced may be at most 20 ppm, or at most 10 ppm; or at most 5 ppm, or at most 2 ppm, or at most 1 ppm. In other aspects of the invention, the impurity gas content of the hydrogen gas produced may be at most 750 parts per billion [ppb] (that is, at most 0.750 ppm); or at most 500 ppb; or at most 200 ppb; or even at most 100 ppb. As known in the art, these impurity contents of the hydrogen gas produced, for example, a content of 1,000 times lower than the impurity gas content of the first gas stream, or an impurity gas content in ppm or in ppb, are typically “on a dry basis.” As known in the art, “on a dry basis” implies that there may be some water vapor in the gas stream produced that has yet to be reduced or removed, for example, in a subsequent drying process.

(21) Another embodiment of the invention is a method for reducing an impurity gas content of a gas stream having a hydrogen gas content and an impurity gas content, the method comprising or including: introducing a first gas stream having a first hydrogen content and a first impurity gas content to a first membrane electrode assembly (MEA) having a first anode containing a catalyst, a first electrolyte, and a first cathode containing a catalyst to produce a second gas stream having a second hydrogen gas content and a second impurity gas content; and passing the second gas stream directly to a second MEA having a second anode containing a catalyst, an second electrolyte, and a second cathode containing a catalyst to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen content and a third impurity gas content less than the first impurity gas content.

(22) In one aspect, the first MEA and the second MEA may be positioned in a hydrogen purifying cell, and passing the second gas stream directly to a second MEA may comprise passing the second gas stream to the second MEA without allowing the second gas stream to leave the hydrogen purifying cell.

(23) In one aspect, the method may further include removing at least some of the second gas stream to produce a modified second gas stream having a reduced non-hydrogen gas partial pressure than the second gas stream, and then introducing the modified second gas stream having a reduced non-hydrogen gas partial pressure to the second MEA. In one aspect, the method may further include introducing at least some hydrogen gas to the second gas stream or to the modified second gas stream.

(24) A further embodiment of the invention is a hydrogen gas purifier cell comprising or including: a membrane electrode assembly (MEA) comprising: an anode positioned to contact a first gas stream having a first hydrogen gas content and a first impurity gas content, the anode containing a catalyst adapted to oxidize at least some of first hydrogen gas content to produce hydrogen ions and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions received from the anode; a dual electrode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the dual cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content and to oxidize at least some of second hydrogen gas content in the second gas

stream to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions received from the dual electrode; and a cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte, the cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content.

(25) In one aspect, the hydrogen gas purifier cell further comprises at least one passage for removing at least some of the second gas stream. For example, the at least one passage for removing at least some of the second gas stream may be the dual electrode, for example, a permeability of the dual electrode; a gas permeable diffusion layer; and/or the second electrolyte.

(26) In one aspect, the hydrogen gas purifier cell further comprises at least one passage for introducing at least some hydrogen gas to the second gas stream. For example, the at least one passage for introducing at least some hydrogen gas may be the dual electrode, for example, a permeability of the dual electrode; a gas permeable diffusion layer; and/or the second electrolyte, for instance, via "back diffusion."

(27) A further embodiment of the invention is a method of purifying hydrogen gas comprising or including: introducing a first gas stream having a first hydrogen gas content and a first impurity gas content to an anode containing a catalyst; in the anode, catalytically oxidizing at least some of first hydrogen gas content to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions produced in the anode through a first electrolyte to a dual electrode; in the dual electrode, catalytically reducing the at least some of the hydrogen ions transferred through the first electrolyte to produce a second gas stream having a second hydrogen gas content, and catalytically oxidizing at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions produced in the dual electrode through a second electrolyte to a cathode, and in the cathode, catalytically reducing the at least some of the hydrogen ions transferred through the second electrolyte to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content.

(28) In one aspect, the method may further include removing at least some of the second gas stream, for example, through at least one passage. For example, the at least one passage for removing at least some of the second gas stream may be a gas diffusion layer; a gas-permeable dual electrode; and/or the second electrolyte, for instance, via "back diffusion."

(29) In one aspect, the method may further include introducing at least some hydrogen gas to the second gas stream, for example, through at least one passage. For example, the at least one passage for introducing at least some hydrogen gas may be through a gas diffusion layer; through a gas-permeable dual electrode; and/or through the second electrolyte.

(30) A still further embodiment of the invention is a water electrolyzer cell comprising or including: a first membrane electrode assembly (MEA) comprising: a first anode positioned to contact a first H₂O-containing fluid stream, the first anode containing a catalyst adapted to oxidize at least some of the H₂O in the first H₂O-containing fluid stream to produce oxygen gas, hydrogen ions, and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the first anode; and a first cathode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the first cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second fluid stream containing hydrogen gas; a second MEA comprising: a second anode positioned to receive the second fluid stream containing hydrogen gas from the first cathode of the first MEA, the second anode containing a catalyst adapted to oxidize at least some of the hydrogen gas to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the second anode; and a second cathode positioned to receive at least some of the hydrogen ions transferred by the second

electrolyte of the second MEA, the second cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third fluid stream containing hydrogen gas.

(31) In one aspect, the electrolyzer cell may further include at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second fluid stream. For example, the passage for discharging the second fluid stream may be located between the first cathode and the second anode. For instance, the at least one passage located between the first cathode and the second anode may be voids between mating surfaces of the first cathode and the second anode; and/or a gas-permeable layer (GDL) and/or flow field insert, between the first cathode and the second anode.

(32) In another aspect, the water electrolyzer cell may further include at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream. For example, the passage for introducing hydrogen gas may be voids between mating surfaces of the first cathode and the second anode; and/or a GDL and/or flow field insert located between the first cathode and the second anode.

(33) Another embodiment of the invention is a method for electrolyzing water, the method comprising or including: introducing a first H_2O -containing fluid stream to a first anode containing a catalyst; in the first anode, catalytically oxidizing at least some of the H_2O in the first H_2O -containing fluid stream to produce oxygen gas, hydrogen ions, and electrons; transferring at least some of the hydrogen ions through a first electrolyte to a first cathode containing a catalyst; in the first cathode, catalytically reducing the at least some of the hydrogen ions transferred through the first electrolyte to produce a second fluid stream having hydrogen gas; introducing the second fluid stream having the hydrogen gas to a second anode having a catalyst; in the second anode, catalytically oxidizing at least some of the hydrogen gas in the second fluid stream to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions produced at the second anode through a second electrolyte to a second cathode; and in the second cathode, catalytically reducing the at least some of the hydrogen ions transferred through the second electrolyte to produce a third fluid stream having hydrogen gas. According to aspects of the invention, the "fluid streams" may be a liquid stream, a gas stream, and/or a liquid and gas stream.

(34) In one aspect, the method may further include removing at least some of the second fluid stream to produce a modified fluid stream having a non-hydrogen gas partial pressure lower than a partial pressure of the non-hydrogen gas in the second fluid stream. The method may further include introducing the modified fluid stream to the second anode.

(35) In another aspect, the method may further include introducing some hydrogen gas to the second fluid stream, for example, the hydrogen gas introduced may replenish at least some of the hydrogen gas removed from the second fluid stream.

(36) A further embodiment of the invention is a water electrolyzer cell comprising or including: a membrane electrode assembly (MEA) comprising: an anode positioned to contact a first H_2O -containing fluid stream, the anode containing a catalyst adapted to oxidize at least some of the H_2O in the first H_2O -containing fluid stream to produce oxygen gas, hydrogen ions, and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the anode; a dual electrode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the dual cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content and to oxidize at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions received from the dual electrode; and a cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte of the second MEA, the cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third fluid stream containing hydrogen gas.

(37) Another embodiment of the invention is a method for electrolyzing water, the method

comprising or including: introducing a first H.sub.2O-containing fluid stream to an anode containing a catalyst; in the anode, catalytically oxidizing at least some of the H.sub.2O in the first H.sub.2O-containing fluid stream to produce oxygen gas, hydrogen ions, and electrons; transferring at least some of the hydrogen ions produced in the anode through a first electrolyte to a dual electrode; in the dual electrode, catalytically reducing the at least some of the hydrogen ions transferred through the first electrolyte to produce a second fluid stream having a second hydrogen gas content, and catalytically oxidizing at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; transferring at least some of the hydrogen ions produced in the dual electrode through a second electrolyte to a cathode; and in the cathode, catalytically reducing the at least some of the hydrogen ions transferred through the second electrolyte to produce a third fluid stream having hydrogen gas.

(38) These and other aspects, features, and advantages of this invention will become apparent from the following detailed description of the various aspects of the invention taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The subject matter, which is regarded as the invention, is particularly pointed out and distinctly claimed in the claims at the conclusion of the specification. The foregoing and other objects, features, and advantages of the invention will be readily understood from the following detailed description of aspects of the invention taken in conjunction with the accompanying drawings in which:

(2) FIG. 1 is a schematic illustration of a hydrogen gas purifier cell according to one aspect of the invention.

(3) FIG. 2 is a schematic illustration of a hydrogen gas purifier cell according to another aspect of the invention.

(4) FIG. 3 is a schematic elevation view of a hydrogen purifier stack assembly having the hydrogen gas purifier cell shown in FIG. 1 or FIG. 2 according to aspects of the invention.

(5) FIG. 3A is a detailed view of the hydrogen purifier stack assembly shown in FIG. 3 as identified by Detail 3A shown in FIG. 3.

(6) FIG. 3B is a detailed view similar to FIG. 3A according to another aspect of the invention.

(7) FIG. 4 is a schematic elevation view of a hydrogen purifier stack having a plurality of a hydrogen purifier cells shown in FIG. 1 and/or FIG. 3, according to an aspect of the invention.

(8) FIG. 5 is a schematic illustration of a hydrogen gas purifier cell according to another aspect of the invention.

(9) FIG. 6 is a schematic view of a hydrogen gas purifier system having one or more hydrogen purifiers having one or more hydrogen purifier stacks shown in FIG. 4, according to an aspect of the invention.

(10) FIG. 7 is a schematic illustration of a water electrolyzer cell according to another aspect of the invention.

(11) FIG. 8 is a schematic illustration of a water electrolyzer cell according to another aspect of the invention.

DETAILED DESCRIPTION OF THE INVENTION

(12) FIG. 1 is a schematic illustration of a hydrogen gas purifier cell **10** according to one aspect of the invention. According to this aspect, purifier cell **10** is positioned and adapted to receive a feed or first gas stream **12** having at least some hydrogen gas content **14** (that is, diatomic hydrogen gas, H.sub.2) and at least some non-hydrogen gases **16** and produce a reduced non-hydrogen-gas-content gas stream **18**, that is, a gas stream **18** having a purer hydrogen gas content. The non-

hydrogen gases **16** may typically contain nitrogen (N.sub.2), argon (Ar), carbon monoxide (CO), methane (CH.sub.4), oxygen (O.sub.2), and/or carbon dioxide (CO.sub.2), among other gases. According to one aspect of the invention, the expression “non-hydrogen gases” may refer to gases that are not diatomic hydrogen gas, H.sub.2. The reduced non-hydrogen-gas-content gas stream **18** may be referred to as a discharge gas stream **18** or a third gas stream **18**. Third gas stream **18** may typically include an enhanced hydrogen gas content **20** and a reduced non-hydrogen gas content **22**, for example, a higher purity hydrogen gas stream, for instance, having a non-hydrogen content of at most 100 ppm, on a dry basis. In other aspects of the invention, the non-hydrogen content **22** of gas stream **18** may be at most 20 ppm, or at most 10 ppm; or at most 5 ppm, or at most 2 ppm, or at most 1 ppm. In addition to reducing the content of the non-hydrogen gases in gas stream **18**, in aspects of the invention, gas stream **18** may typically have increased hydrogen gas content **20**, for example, by volume percent, and an increased hydrogen gas pressure, for example, a pressure greater than the pressure of the feed gas stream **12**. In one aspect, the pressure of gas stream **18** may be less than the pressure of gas stream **12**. Since the desired function of purifier cell **10** is to reduce or substantially eliminate the content of non-hydrogen gases **16**, the non-hydrogen gases **16** may be referred to as “impurity gases” **16** or “first impurity gases” **16**.

(13) As shown schematically in FIG. 1, hydrogen gas purifier cell **10** typically comprises a multilayer structure having components, for example, anodes and cathodes having thin planar or thin laminar construction, where the structures shown in FIG. 1 may comprise a side elevation view or a trans-axial cross-sectional view of purifier cell **10** that is not drawn to scale, but is drawn to facilitate disclosure of the invention.

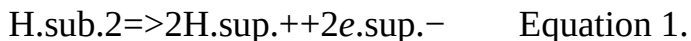
(14) According to aspects of the invention, to provide the desired increased hydrogen gas content **20**, and typically increased hydrogen gas pressure, purifier cell **10** typically includes a first membrane electrode assembly (MEA) **24** and at least one second MEA **26**. The first MEA **24** includes a first electrode **28**, specifically a first “anode” **28**, as referred to in the art. Anode **28**, any anode disclosed herein, may typically be gas-permeable, specifically, hydrogen-gas permeable, where at least some of the hydrogen gas content **14** and at least some of the non-hydrogen gas content **16** in first gas stream **12** may pass through anode **28**, for example, in an axial direction as indicated by the arrow of first gas stream **12**. In addition, anode **28**, and any anode or cathode disclosed herein, includes at least some catalyst, for example, at least some platinum group metal-containing catalyst, capable of enhancing the oxidation of hydrogen to hydrogen ions (H.sup.+), for instance, a platinum-containing catalyst, though in some aspects, a non-platinum group metal-containing catalyst may be used for anode **28** and any anodes or cathodes disclosed herein. As known in the art, a platinum group metal-containing catalyst may be a catalyst containing at least some nickel (Ni), at least some palladium (Pd), and/or at least some platinum (Pt).

(15) The first anode **28** is positioned to contact the first gas stream **12** having the first hydrogen gas content **14** and the first impurity gas content **16**. The relative content of the first hydrogen gas **14** and the first impurity gas content **16** of first gas stream **12** is illustrated schematically in FIG. 1, and in other figures, by the partial shading of the arrow identified as first gas stream **12**. This partial shading of gas stream arrow **12** (and of gas stream arrows **38** and **18** in FIG. 1) are for illustration only, and do not represent actual relative gas contents of these gas streams according to aspects of the invention.

(16) In one aspect, in order to enhance the distribution of first gas stream **12** about the surface of anode **28**, an electrically-conductive gas diffusion layer (GDL), not shown in FIG. 1, may be positioned between first gas stream **12** and anode **28**, for example, the GDL may be applied over the surface of anode **28** contacted by first gas stream **12**. In one aspect, the GDL used for the cell **10**, or the GDL layer used in any aspect disclosed herein, may be a carbon fiber-type GDL, for example, one provided by SGL Carbon GmbH, or its equivalent. In another aspect, a gas-distribution or flow field insert (as disclosed herein) with or without a GDL, may be positioned over anode **102** to promote or enhance the distribution of the first gas stream **12** across the surface

of anode **28**.

(17) According to aspects of the invention, the catalyst contained in first anode **28** promotes or enhances the oxidation of at least the hydrogen gas (H.sub.2) content **14** introduced to anode **28** to yield or produce hydrogen ions (H.sup.+) and electrons (e.sup.-) pursuant to Equation 1, as known in the art.



Due to the permeability of anode **28**, hydrogen gas (H.sub.2) passes into anode **28**, and, due to the electrical conductivity of anode **28**, electrons (e.sup.-) are conducted away from anode **28**, and, according to aspects of the invention, hydrogen ions (H.sup.+) are introduced to electrolyte **30**. As known in the art, hydrogen ions (H.sup.+) are protons. However, it is recognized in the art that at least some undesirable, non-hydrogen gas **16** also passes through anode **28**.

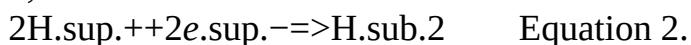
(18) As is typical in the art, at least some of input or first gas stream **12** may not oxidize at anode **28**, but be removed as gas stream **13**, for example, an “exhaust gas stream.” Typically, exhaust stream **13**, typically having less hydrogen gas content due to the oxidation of hydrogen gas that occurs in first anode **28**, may be captured and directed, for example, via channels, manifolds, and ports, to further processing or disposed of as needed.

(19) Electrolyte **30**, or a first electrolyte **30**, is positioned and adapted to receive and transfer at least some of the hydrogen ions (H.sup.+) . Due to the close proximity of anode **28** to electrolyte **30**, hydrogen ions (H.sup.+) and non-hydrogen gas that diffused through anode **28** are passed from anode **28** to electrolyte **30**.

(20) First electrolyte **30** comprises a barrier between the first anode **28** and the electrode **32**. First electrolyte **30** may comprise any material or substance capable of transmitting the hydrogen ions (H.sup.+), for example, selectively transmitting hydrogen ions (H.sup.+), that is, protons, from first anode **28** to electrode **32**. That is, in one aspect, electrolyte **30**, and any electrolyte disclosed herein, may be referred to as a “proton-conductive material,” while substantially preventing the flow of gas and electrons. First electrolyte **30**, and any electrolyte disclosed herein, may typically be acidic, for example, an acidic polymer containing a perfluorosulfonic acid (PFSA). In one aspect, electrolyte **30**, and any electrolyte disclosed herein, may be a membrane marketed under the trademark Nafion™ by The Chemours Company of Wilmington, Delaware, or its equivalent. In other aspects, electrolyte **30**, and any electrolyte disclosed herein, may contain one or more of the following acids: phosphoric acid [H.sub.3PO.sub.4], sulfuric acid [H.sub.2SO.sub.4], or any other hydrogen ion (H.sup.+) conducting acid. In one aspect, first electrolyte **30** may comprise a proton exchange membrane (PEM), as known in the art.

(21) As known in the art, the passage of gas through electrolyte **30**, and through any electrolyte disclosed herein, is driven by the partial pressure gradient of the gas across the electrolyte, for example, from one side of electrolyte **30** to the other, opposite side of electrolyte **30**. Thus, any undesirable non-hydrogen gas with sufficient partial pressure gradient may also diffuse through electrolyte **30**, and through any electrolyte disclosed herein. In addition to the pressure gradient across an electrolyte, imperfections in the electrolyte, for example, small holes or voids in the electrolyte, may also undesirably allow gas to flow through an electrolyte, like electrolyte **30**.

(22) Electrode **32**, which may be referred to as “cathode” **32**, as known in the art, or a first cathode **32**, is positioned to receive at least some of the hydrogen ions (H.sup.+) transferred by the first electrolyte **30**. Similar to the first anode **28**, first cathode **32**, and any cathode disclosed herein, typically contains a catalyst, for example, a platinum group metal-containing catalyst, adapted to enhance the reaction (that is, reduction) of at least some of the hydrogen ions (H.sup.+) that pass through first electrolyte **30** by reacting with at least some electrons (e.sup.-) pursuant to Equation 2, as known in the art.



(23) The resulting, or “evolved,” hydrogen gas (H.sub.2) **34**, or a second hydrogen gas content **34**, and any non-hydrogen gases **36**, or a second impurity gas content **36**, transferred through first

cathode **32** are indicated as gas stream **38** in FIG. **1** or a second gas stream **38**.

(24) In one aspect, in order to enhance the distribution of the hydrogen gas about the surface of first cathode **32**, a gas diffusion layer (GDL) and/or a flow field insert, not shown in FIG. **1**, may be positioned between first electrolyte **30** and first cathode **32**, for example, the GDL and/or flow field insert may be applied over the surface of first cathode **32**.

(25) According to aspects of the invention, the second hydrogen gas content **34** is greater than the first hydrogen gas content **14**; and the second impurity gas content **36** is less than the first impurity gas content **16**.

(26) As shown in FIG. **1**, according to aspects of the invention, second gas stream **38** having second hydrogen gas content **34** and second impurity gas content **36** may then be introduced to second MEA **26**, specifically, to electrode **40**, or second anode **40**, of second MEA **26**. In the schematic diagram of the purifier cell **10** shown in FIG. **1**, first MEA **24** is shown spaced from second MEA **26** to facilitate illustration and disclosure of the invention. However, according to aspects of the invention, the spacing between first MEA **24** and second MEA **26** may be minimal, for example, where the surface of first cathode **32** may abut or contact the surface of second anode **40**. However, in one aspect, at least some spacing, for example, 0.1 millimeters [mm] to 0.5 mm, may be present between the surface of first cathode **32** and the surface of second anode **40**.

(27) According to aspects of the invention, and contrary to the existing art, second gas stream **38** having second hydrogen gas content **34** and second impurity gas content **36** may then be introduced to second MEA **26** without removing or extracting second gas stream **38** from purifier cell **10**. In other words, though some portion of gas stream **38** may undesirably “escape” from purifier cell **10**, according to one aspect of the invention, substantially all of gas stream **38** produced at first cathode **32** or diffused through cathode **32** is received by second anode **40**. For example, in one aspect, second gas stream **38** may be allowed to pass from first cathode **32** of MEA **24** to second anode **40** of MEA **26** without any intervening handling or treatment, for instance, without passing outside of cell **10**. In one aspect, the evolved hydrogen gas (H.sub.2) formed in first cathode **32** may substantially be immediately oxidized to hydrogen ions (H.sup.+) at second anode **40**. In one aspect, second gas stream **38** may be allowed to pass directly from first cathode **32** of MEA **24** to second anode **40** of MEA **26** of purifier cell **10**, for instance, without passing outside of cell **10** before reaching second cathode **40**.

(28) The second anode **40** is positioned to be contacted by the second gas stream **38** having the second hydrogen gas content **34** and the second impurity gas content **36**. Second anode **40** of second MEA **26** may be similar to, if not identical to, first anode **28** of MEA **24**. Second anode **40** may be hydrogen-gas permeable, where at least some of the hydrogen gas content **34** and at least some of the non-hydrogen gas content **36** in second gas stream **38** may pass through second anode **40**, for example, in an axial direction, as indicated by the arrow of second gas stream **38**. In addition, second anode **40** includes at least some catalyst, for example, at least some platinum group metal-containing catalyst, capable of enhancing the oxidation of hydrogen gas to hydrogen ions (H.sup.+) and electrons (e.sup.-), as indicated by Equation 1.

(29) In one aspect, in order to enhance the distribution of second gas stream **38** about the surface of second anode **40**, a gas diffusion layer or GDL and/or flow field insert, not shown in FIG. **1**, may be positioned between the first cathode **32** and second anode **40**, for example, the GDL and/or flow field insert may be applied over the surface of second anode **40** contacted by second gas stream **38**.

(30) According to aspects of the invention, the catalyst contained in second anode **40** promotes or enhances the oxidation of the hydrogen gas content **34** introduced to second anode **40** to yield or produce hydrogen ions (H.sup.+) and electrons (e.sup.-) pursuant to Equation 1, above. In one aspect, the electrons (e.sup.-) generated in second anode **40** may be directed back to first cathode **32**, as indicated by arrow **41** in FIG. **1**, to supply at least some of the electrons (e.sup.-) used to reduce the hydrogen ions (H.sup.+) to hydrogen (H.sub.2) at first cathode **32**, pursuant to Equation 2. Due to the permeability of second anode **40**, some impurity gases and any unoxidized hydrogen

(H.sub.2) may pass through second anode **40**, and, according to aspects of the invention, are introduced to or come in contact with an electrolyte **42**.

(31) Electrolyte **42** of MEA **26**, or a second electrolyte **42**, is positioned and adapted to receive and transfer at least some of the hydrogen ions (H.sup.+) received from the second anode **40**. Second electrolyte **42** comprises a gas barrier between the second anode **40** and the electrode **44**. Second electrolyte **42** may be similar to, if not substantially identical to, first electrolyte **30**, and comprise any material or substance capable of transmitting the hydrogen ions (H.sup.+), for example, selectively transmitting hydrogen ions (H.sup.+), that is, protons, from second anode **40** to an electrode **44**. Again, as noted with respect to electrolyte **30**, in one aspect, electrolyte **42** may be referred to as a “proton-conductive material.” Second electrolyte **42** may typically be acidic, for example, containing one or more of the acids identified above with respect to first electrolyte **30**. However, in one aspect, second electrolyte **42** may comprise a PEM, as known in the art.

(32) Electrode **44**, which may be referred to as “cathode” **44**, as known in the art, or second cathode **44**, is positioned to receive at least some of the hydrogen ions (H.sup.+) transferred by second electrolyte **42**. Similar to the second anode **40**, second cathode **44** typically contains a catalyst, for example, a platinum group metal-containing catalyst, adapted to enhance the reduction of at least some of the hydrogen ions (H.sup.+) using at least some electrons (e.sup.-) pursuant to Equation 2, above. The resulting, or “evolved,” hydrogen gas (H.sub.2) content **20**, or a third hydrogen gas content **20**, and any non-hydrogen gases **22**, or a second impurity gas content **22**, transferred through electrolyte **30** are indicated as gas stream **18** in FIG. **1**, or discharge gas stream **18** or third gas stream **18**.

(33) In one aspect, in order to enhance the distribution of the hydrogen ions (H.sup.+) about the surface of second cathode **44**, a gas diffusion layer (GDL) and/or flow field insert, not shown in FIG. **1**, may be positioned between second electrolyte **42** and second cathode **44**, for example, the GDL and/or flow field insert may be applied over the surface of second cathode **44**.

(34) According to aspects of the invention, the third hydrogen gas content **20** of third gas stream **18** is greater than the first hydrogen gas content **14** and the second hydrogen gas content **34**; and the third impurity gas content **22** is less than the first impurity gas content **16** and the second impurity gas content **36**. However, typically, the third gas stream **18** may have a purity, on a “dry basis,” in percent that is greater than the purity of the first gas stream **12**. For example, the third hydrogen gas content **20** in third gas stream **18** may be at least 10 percent greater, by volume, than the first hydrogen gas content **14** in first gas stream **12**. In one aspect, the third hydrogen gas content **20** may be from 20 percent to 30 percent greater, by volume, than the first hydrogen gas content **14**. Also, in one aspect, the third gas stream **18** may be at least 1,000 times purer in hydrogen, by volume, than the first gas stream **12**. In one aspect, the third gas stream **18** may be from 10,000 times to 10,000,000 [10 million] times purer in hydrogen, by volume, than the first gas stream **12**. However, typically, the third gas stream **18** may be from 100,000 times to 2,000,000 [2 million] times purer in hydrogen content, by volume, than the first gas stream **12**. For example, in one aspect, the purity of the third gas stream **18** may be at least 99.99 percent hydrogen, by volume, or at least 99.999 (five 9s) percent, by volume, or 99.9999 (six 9s) percent by volume. According to another aspect of the invention, the purity of the third gas stream **18** may be expressed on a basis of the third impurity gas content **22**. For example, in one aspect the third impurity gas content **22** may be at most 100 ppm of impurity gas. In other aspects of the invention, the third impurity gas content **22** produced may be at most 20 ppm, or at most 10 ppm, at most 5 ppm, or at most 2 ppm, or at most 1 ppm. In other aspects of the invention, the third impurity gas content **22** of the hydrogen gas **18** produced may be at most 750 ppb (that is, at most 0.750 ppm), or at most 500 ppb, or at most 200 ppb; or even at most 100 ppb. As known in the art, these impurity contents of the hydrogen gas produced are typically “on a dry basis.”

(35) According to aspects of the invention, in purifier cell **10**, by passing the second gas stream **38** having hydrogen gas content **34** from first cathode **32** of MEA **24** to second anode **40** of second

MEA **26** can not only provide a more compact purifier, but also can provide a more efficient device and providing a purer hydrogen gas content **22** than the prior art. Among other things, the passing of the second gas stream **38** from first cathode **32** to second anode **40**, for example, directly, avoids the recognized addition of components, the losses in efficiency, and loss in hydrogen gas content that characterize the performance of prior art hydrogen gas purifiers.

(36) In addition, according to aspects of the invention, the resulting gas stream, that is, third gas stream **18** having a higher hydrogen gas content **20** may typically be provided at a higher pressure than the pressure of the feed or first gas stream **12**. For example, according to aspects of the invention, where first gas stream **12** may have a pressure of about 1 pound per square inch-gauge [psig], third gas stream **18** may have a pressure of at least 150 psig. In one aspect, the pressure of third gas stream **18** may be at least 120 psig, or at least 200 psig, or even at least 10,000 [10 thousand] psig. In other aspects of the invention, the pressure of third gas stream **18** may not be greater than the pressure of the feed or first gas stream **12**; in one aspect, the pressure of the third gas stream **18** may less than the pressure of the first gas stream **12**.

(37) FIG. **2** is a schematic illustration of a hydrogen gas purifier cell **50** according to another aspect of the invention. According to this aspect, purifier cell **50** may have many of the features of purifier cell **10**; however, purifier cell **50** further includes at least one gas discharge or release between the MEAs. According to this aspect, testing has shown that that discharging or allowing at least some of the second gas stream (**38** in FIG. **2**) to escape from purifier cell **50** yields a higher hydrogen gas content, that is, compared with a purifier not allowing the removal of at least some of the second gas stream between MEAs. It is believed that this venting of at least some of the second gas stream reduces the partial pressure of the non-hydrogen gas between the MEAs and thus reduces the partial pressure gradient driving force of the undesirable non-hydrogen gas through the second MEA. In a further aspect, since some of the desirable hydrogen gas will be lost when removing some of the second gas stream, at least some hydrogen gas may be introduced between the MEAs to function as a “make-up” hydrogen gas for the hydrogen gas possibly lost with the removal of some of the second gas stream.

(38) As shown in FIG. **2**, in a manner similar to purifier cell **10**, purifier cell **50** is positioned and adapted to receive a feed or first gas stream **12**, similar or identical to first gas stream **12** shown in FIG. **1**, and at least some non-hydrogen gases **16** and produce an increased hydrogen gas content and a reduced non-hydrogen-gas-content gas stream **52**. The gas stream **52** may be referred to a discharge gas stream **52** or a third gas stream **52**. Similar to purifier cell **10**, third gas stream **52** may typically include an enhanced hydrogen gas content **54** and a reduced non-hydrogen gas content **56**. As shown schematically in FIG. **2**, similar to purifier cell **10**, hydrogen gas purifier cell **50** typically comprises a multilayer structure having components, for example, anodes and cathodes having thin planar or thin laminar construction, where the structures shown in FIG. **2** may comprise a side elevation view or a trans-axial cross-sectional view of purifier cell **50** that is not drawn to scale, but is drawn to facilitate disclosure of the invention.

(39) In one aspect, hydrogen gas purifier cell **50** shown in FIG. **2**, may have substantially the same first MEA **24** as in purifier cell **10**, that is, having a first anode **28**, a first electrolyte **30**, and a first cathode **32** to at least partially purify first gas stream **12** and yield a second gas stream **38** having a hydrogen gas content **34** and a non-hydrogen gas content **36** in a manner substantially identical to purifier cell **10** shown in FIG. **1**. As is typical in the art, at least some of input or first gas stream **12** may not diffuse through anode **28**, but be removed as gas stream **13**, for example, an “exhaust gas stream.” In addition, hydrogen gas purifier cell **50** may have substantially the same second MEA **26** having a second anode **40**, a second electrolyte **42**, and a second cathode **44** to at least partially purify gas stream **38** to yield a third gas stream **52** having a hydrogen gas content **54** and a non-hydrogen gas content **56**. However, according to the aspect of the invention shown in FIG. **2**, hydrogen gas purifier cell **50** includes at least one discharge gas stream or displaced gas stream **58** having a non-hydrogen gas content **60** and a hydrogen gas content **62**. It is recognized that the non-

hydrogen gas content **60** and the hydrogen gas content **62** of discharge gas stream **58** may be substantially the same as the non-hydrogen gas content **36** and the hydrogen gas content **34** of second gas stream **38**.

(40) In one aspect, in order to enhance the distribution of the gas streams about the electrode surfaces, one or more gas diffusion layers (GDL) and/or flow field insert, not shown in FIG. 2, may be positioned within cell **50**. For example, cell **50** may include a GDL and/or flow field insert associated with first anode **28**, first cathode **32**, second anode **40**, and/or second cathode **44**.

(41) According to this aspect, after first gas stream **12** is processed by first MEA **24** to produce second gas stream **38** having hydrogen gas content **34** and non-hydrogen gas content **36**, at least some of the gas stream **38** is removed via gas stream **58**. The removal of gas stream **58** from gas stream **38** yields a modified or intermediate gas stream **64** having a hydrogen gas content **66** and a non-hydrogen gas content **68**. It is recognized that the non-hydrogen gas content **68** and the hydrogen gas content **66** of modified gas stream **64** may be substantially the same as the non-hydrogen gas content **36** and the hydrogen gas content **34** of second gas stream **38**. According to aspects of the invention, removal of gas stream **58** reduces partial pressure of non-hydrogen gas content **68** in modified gas stream **64** and this reduction in partial pressure reduces the partial pressure gradient across the second MEA **26** of the non-hydrogen gas content **58** which reduces the passage of the non-hydrogen gas content **68** through second MEA **26** to third gas stream **52**. Accordingly, according to aspects of the invention, the non-hydrogen gas content **56** of third gas stream **52** is reduced.

(42) The removal of gas stream **58** with non-hydrogen content **60** may be practiced by various means. In one aspect, gas stream **58** may be removed by simply venting at least some of second gas stream **38**, for example, venting through the inherent spacing between first MEA **24** and second MEA **26**, for instance, via the inherent spaces or voids between the surface of first cathode **32** and second anode **40**. In another aspect, gas stream **58** may be removed by providing paths, channels, or grooves, for example, radial or transverse channels or grooves, in the mating surface of first cathode **32**, in the mating surface of second anode **40**, or in both the mating surface of first cathode **32** and in the mating surface of second anode **40**. In another aspect, gas stream **58** may be removed through a GDL and/or a flow field insert positioned between the mating surface of first cathode **32** and the mating surface of second anode **40**. As known in the art, a GDL is typically a porous material, such as, carbon paper, through which gas stream **58** may pass. In another aspect, gas stream **58** may be removed by providing one or more spacers between the mating surface of first cathode **32** and the mating surface of second anode **40** to provide a pathway for gas stream **58**. In one aspect, a source of vacuum may be introduced to draw out at least some of the second gas stream **38** via gas stream **58**.

(43) According to another aspect of the invention, after first gas stream **12** is processed by first MEA **24** to produce second gas stream **38**, having hydrogen gas content **34** and non-hydrogen gas content **36**, at least some hydrogen gas may be introduced to gas stream **38**. As shown in FIG. 2, in one aspect, hydrogen gas may be introduced to second gas stream **38** via gas stream **59** (shown in phantom in FIG. 2) to replace the hydrogen gas **62** lost from second gas stream **38** via gas stream **58** and produce modified gas stream **64**. Though in one aspect, gas stream **38** may be highly pure hydrogen gas, for example, having a purity to at least greater than the hydrogen content **14** of first gas stream **12**; in other aspects, gas stream **59** may be a hydrogen-gas containing stream having at least some hydrogen gas content, but may have a non-hydrogen gas content.

(44) The introduction of hydrogen-containing gas stream **59**, for example, a “make-up gas stream,” may be practiced with or without the removal of gas stream **58**. The introduction of gas stream **59** to gas stream **38** may be practiced in any one or more convenient manner, for example, by introducing the hydrogen-gas containing stream **59** through the gas-permeable electrolyte **42**, for example, driven by a hydrogen gas partial pressure gradient, or through the gas-permeable first cathode **32**, through a gas permeable GDL and/or flow field insert, or through channels in first

cathode **32**, channels in second anode **40** or through channels in both first cathode **32** and in second anode **40**. Any channels that may be provided for hydrogen-gas containing stream **59** may be located in one or both of the opposing surfaces of first cathode **32** and second anode **40**, that is, the surfaces the space occupied by second gas stream **38**. In one aspect, the make-up hydrogen gas stream **59** may comprise at least some of the third gas stream **52** having a third hydrogen content **54**. For example, at least some of the third gas stream **52** may be introduced to the second gas stream **38** by diffusion through the second electrolyte **42**, as indicated in phantom by gas stream **59A** in FIG. **2**. This diffusion through the second electrolyte **42** may be referred to as “back diffusion” of at least some of the third gas stream **52** having the third hydrogen content **54** through the second electrolyte **42** to provide at least some of the make-up gas stream **59** to the second gas stream **38** or to the modified gas stream **64**. The make-up gas stream **59** may be provided any one or more of these mechanisms.

(45) According to aspects of the invention, the third hydrogen gas content **54** of third gas stream **52** is greater than the first hydrogen gas content **14** and the second hydrogen gas content **34**; and the third impurity gas content **56** is less than the first impurity gas content **16** and the second impurity gas content **36**. However, typically, the third gas stream **52** may have a purity, on a “dry basis,” in percent that is greater than the purity of the first gas stream **12**. For example, in one aspect, the purity of the third hydrogen gas content **54** may be at least 99.99 percent, by volume, or at least 99.999 (five 9s) percent, by volume, or 99.9999 (six 9s) percent, by volume. According to another aspect of the invention, the purity of the third gas stream **52** may be expressed on a basis of the third impurity gas content **56**. For example, in one aspect, the third impurity gas content **56** may be at most 100 ppm of impurity gas. In other aspects of the invention, the third impurity gas content **56** produced may be at most 20 ppm, or at most 10 ppm, or at most 5 ppm, or at most 2 ppm, or at most 1 ppm. In other aspects of the invention, the third impurity gas content **56** of the hydrogen gas produced may be at most 750 ppb (that is, at most 0.750 ppm); or at most 500 ppb; or at most 200 ppb; or even at most 100 ppb. As known in the art, these impurity contents of the hydrogen gas produced are typically “on a dry basis.”

(46) FIG. **3** is a schematic elevation view of a hydrogen purifier stack assembly **80** having a hydrogen gas purifier cell **82**, for example, hydrogen gas purifier cell **10** or cell **50** disclosed herein, positioned between opposing electrically conductive, but gas impermeable layers or plates **84** and **86** and electrically conductive layers or bus bars **88** and **90** according to an aspect of the invention. Electrically conductive, but gas impermeable layers or plates **84** and **86** may be referred to as “bi-polar plates” as known in the art, since plates **84** and **86** may typically contain passages or channels adapted to introduce or remove gases from cell **82**. Electrically conductive layers or bus bars **88** and **90** may also be referred as “current collectors.” As shown, in one aspect, cell **82** may include at least two MEAs, a first MEA **92** and at least a second MEA **94**. However, it is envisioned that, according to aspects of the invention, cell **82** may include three or more MEAs **92**, **94**; or 5 or more MEAs **92**, **94**, for example, positioned between bus bars **88** and **90**. In one aspect, at least 10 MEAs **92**, **94** may be positioned between bus bars **88** and **90**.

(47) In one aspect, hydrogen gas purifier cell **82** may be referred to as a “double membrane electrode assembly” or “DMEA.” First MEA **92** includes a first anode **96**, a first electrolyte **98**, and a first cathode **90**. First anode **96** may be similar, if not identical, to first anode **28** disclosed herein; first electrolyte **98** may be similar, if not identical, to first electrolyte **30** disclosed herein; and first cathode **90** may be similar, if not identical, to first cathode **32** disclosed herein. Second MEA **94** includes a second anode **102**, a second electrolyte **104**, and a second cathode **106**. Second anode **102** may be similar, if not identical, to second anode **40** disclosed herein; second electrolyte **104** may be similar, if not identical, to second electrolyte **42** disclosed herein; and second cathode **106** may be similar, if not identical, to second cathode **44** disclosed herein.

(48) Electrically conductive, gas-impermeable layers or plates (or bi-polar plates) **84** and **86** may typically be fabricated from a corrosion resistant or non-oxidative material due to the electro-

chemistry of cell **82**. Though it is envisioned that any electrically-conductive, substantially gas-impermeable, and substantially corrosion-resistant material, for example, a metal, may be used in aspects of the invention for plates **84** and **86**, typically, plates **84** and **86** may be made from a electrically-conductive, substantially gas-impermeable, and substantially corrosion resistant graphite-containing material. For example, plates **84** and **86** may be made from material containing graphite powder and a resin. In one aspect, bi-polar plates **84** and **86** may be made from a non-metallic material, for example, a plastic, into which electrically conductive inserts and/or particles have been placed to provide the desired conductivity.

(49) In one aspect, it is envisioned that bi-polar plates **84** and **86** may include portions that are at least semi-permeable, for example, to allow for fluid permeability as needed for thermal management and/or water management.

(50) According to an aspect of the invention, electrically conductive current collectors **88** and **90** may be relatively highly conductive, for example, relative to plates **84** and **86**. In one aspect, current collectors **88** and **90** may be plated, for example, gold or silver plated to enhance conductivity. For example, current collectors **88** and **90** may comprise stainless steel plates, aluminum plates, or copper plates that may be plated with gold or silver.

(51) According to aspects of the invention, as shown in FIG. 3, a first gas stream **108** containing hydrogen gas (H.sub.2) and non-hydrogen gas, as disclosed herein, may be introduced to first anode **96** where at least some of the hydrogen gas (H.sub.2) oxidizes to hydrogen ions (H⁺) and electrons (e⁻) pursuant to Equation 1. The first gas stream **108** may be introduced to first anode **96** by any conventional means, for example, via passages and/or channels in bi-polar **84**. For example, as shown in FIG. 3, first gas stream **108** may be introduced to first anode **96** via a plurality of lateral passages **110** communicating with a plurality of longitudinal passages or channels **112** that discharge onto first anode **96**. First anode **96** may include a GDL and/or flow field insert (not shown in FIG. 3), for example, a carbon-paper-type GDL, to enhance the distribution of first gas stream **108** across the surface of first anode **96**.

(52) Once introduced to anode **96**, the reactions and fluid flows described with respect to FIGS. 1 and 2 are generated within MEAs **92** and **94**; specifically, a second gas stream **114** (not shown in FIG. 3) from first cathode **100** and a third gas stream **116** from second cathode **106** are generated according to aspects of the invention. As disclosed herein, second anode **102** receives the second gas stream **114** (again, not shown in FIG. 3). In one aspect, second anode **102** may include or be accompanied by a GDL and/or flow field insert (not shown in FIG. 3), for example, a carbon-paper-type GDL, to enhance the distribution of the second gas stream **114** across the surface of second anode **102**. Third gas stream **116** may be collected from second cathode **106** by conventional means. For example, as shown in FIG. 3, third gas stream **116** may be removed from third cathode **106**, with or without passing through a gas-diffusion layer (not shown), via a plurality of lateral passages **118** communicating with a plurality of longitudinal passages or channels **120** in fluid communication with second cathode **106**. The plurality of lateral passages **118** may comprise a plurality of substantially parallel passages or a plurality of serpentine passages in bi-polar plate **86**. As known in the art, the parallel or serpentine passages in bi-polar plate **86** may be in fluid communication with one or more manifolds, for example, vertical or longitudinal manifolds, and the one or more manifolds may be in fluid communication with one or more ports for introducing, discharging, or re-directing gas streams.

(53) According to aspects of the invention, third gas stream **116** may contain a larger content of hydrogen gas (H.sub.2) and a lower non-hydrogen gas content than the first gas stream **108**. For example, as disclosed herein, the non-hydrogen gas content of third gas stream **116** may have at most 100 ppm of non-hydrogen gas, or at most 20 ppm, or at most 10 ppm, or at most 5 ppm, or at most 2 ppm, or at most 1 ppm, or even at most 500 ppb, "on a dry basis." In addition, third gas stream **116** may typically have a higher pressure than the pressure first gas stream **108**, though in other aspects, third gas stream **116** may have a lower pressure than the pressure first gas stream

108.

(54) A schematic detailed view of the interface between first cathode **100** and second anode **102** according to one aspect of the invention is shown FIG. **3A**. FIG. **3A** is a detailed view of a portion of hydrogen purifier stack assembly **80** shown in FIG. **3** as identified by Detail **3A** shown in FIG. **3**. As shown in FIG. **3A**, second gas stream **114**, having hydrogen gas content **34** (see FIGS. **1** and **2**) and non-hydrogen gas content **36**, exits first cathode **100** and enters spaces **101** between the mating surface of first cathode **100** and second anode **102**. Spaces, voids, or interstitial spaces **101** are typically present between first cathode **100** and second anode **102** due to imperfections in the mating surfaces of first cathode **100** and second anode **102** and due to, among other things, fabrication imperfections and/or tolerances. It is believed that these spaces or voids **101** may be minute, but are typically present between the mating surfaces of first cathode **100** and second anode **102**, and other mating electrodes disclosed herein, and can provide a path for gas streams, for example, the removal and/or introduction of gas streams. According to aspects of the invention, second gas stream **114** via direct contact or via spaces **101** typically contacts and enters second anode **102** where, according to aspects of the invention described herein, catalytic oxidation of the hydrogen gas content **34** in second gas stream **114** occurs. In one aspect, a GDL **105** may be positioned between first cathode **100** and second anode **102**, as shown in FIG. **3B**, to promote or enhance the distribution of the second gas stream **114** across the surface of second anode **102**. In another aspect, a gas-distribution or flow field insert, with or without a GDL, may be positioned between first cathode **100** and second anode **102** (or between any electrodes disclosed herein), as indicated by GDL **105** in FIG. **3B**, to promote or enhance the distribution of the second gas stream **114** across the surface of second anode **102**. The flow field insert may be an electrically-conductive, porous or perforated plate, for example, a porous or perforated metallic plate, or a screen-like insert, for example, a metallic screen-like insert, positioned and adapted to provide at least some gas distribution about the surface of second anode **102**. The flow field insert disclosed herein may also include channels or passages for permitting introduction or removal of gas streams from or to adjacent electrodes.

(55) As disclosed herein, gas purifier cell **82** may comprise hydrogen gas purifier cell **10**, disclosed and describe with respect to FIG. **1**, or hydrogen gas purifier cell **50**, disclosed and describe with respect to FIG. **2**. The detailed views shown in FIGS. **3A** and **3B** also illustrate these aspects of the invention. Specifically, if gas purifier cell **82** in FIG. **3** comprises a purifier cell **10**, as shown in FIG. **3A**, substantially all of second gas stream **114** that leaves first cathode **100** passes, for example, directly passes, through direct contact or via spaces **101** to second anode **102** (with or without GDL **105** and/or flow field insert) for subsequent catalytic oxidation, as disclosed herein.

(56) In aspects of the invention where gas purifier cell **82** in FIG. **3** comprises a purifier cell **50**, as shown in FIGS. **3A** and **3B**, at least some of the second gas stream **114** that leaves first cathode **100** is removed as gas stream **58** (see FIG. **2**). According to this aspect, removed gas stream **58** may contain at least some non-hydrogen gas, where, upon removal, the remaining modified gas stream has a lower partial pressure for the non-hydrogen gas. The lower partial pressure of the non-hydrogen gas is less prone to be transmitted through second electrolyte **104**, and to output or third gas stream **116** (see FIG. **3**). As shown in FIG. **3A**, the removed or vented gas stream **58** may be passed from spaces **101** and/or, as shown in FIG. **3B**, the removed or vented gas stream **58** may be passed through GDL **105** and/or flow field insert.

(57) As also shown in FIGS. **3A** and **3B**, in one aspect, a hydrogen-containing gas stream **59** (shown in phantom in FIGS. **3A** and **3B**) and/or **59A** may also be introduced to second gas stream **114**, with or without the removal of gas stream **58**. For example, as disclosed herein, hydrogen-containing gas stream **59** and/or **59A** may be provided to enhance the hydrogen gas content of the second gas stream **114** introduced to second anode **102**, for example, to enhance the hydrogen content in third gas stream **116** (see FIG. **3**), that is, output stream **116**, and/or to replace at least some of the hydrogen gas content removed in gas stream **58**. As shown in FIGS. **3A** and **3B**, the

introduction of hydrogen-containing gas stream **59** and/or **59A** into space **101** may be practiced with or with the presence of GDL **105**, for example, as shown in FIG. **3B**, hydrogen-containing gas stream **59** may be introduced through a gas-permeable GDL **105** and/or flow field insert.

(58) FIG. **4** is a schematic elevation view of a hydrogen purifier stack **130** having a plurality of a hydrogen purifier cells **132A** through **132N**, for example, having two or more hydrogen purifier cells **10** shown in FIG. **1** and/or two or more hydrogen cells **50** shown in FIG. **2**, according to an aspect of the invention. According to aspects of the invention, “N” is the number of purifier cells that may be included in hydrogen purifier stack **130** according to aspects of the invention.

Specifically, it is envisioned that N may range from 1 to 1,000 cells, but typically may range from 40 cells to 100, for example, 80 cells.

(59) According to this aspect of the invention, each of the hydrogen purifier cells **132A-132N** include double MEAs (DMEAs), as disclosed herein, separated by electrically-conductive and gas-impermeable layers or plates **134A-134N+1**, for example, “bi-polar plates.” Layers or plates **134A-134N+1** may be similar to and have the same characteristics as layers or plates **84** and **86** shown and described with respect to FIG. **3**. For example, layers or plates **134A-134N+1** may have one or more lateral passages and a plurality of axial passages or channels, for example, flow distribution passages, adapted and located to introduce gas flows to hydrogen purifier cells **132A-132N** and/or remove gas flows from hydrogen purifier cells **132A-132N**. According to aspects of the invention, some plates **134A-134N+1**, for example, plate **134B** shown in FIG. **4**, may include passages, for example, isolated separate passages, adapted and located to both remove a gas stream from a second cathode of a DMEA **132A** and introduce a gas stream from a first anode of a DMEA **132B**. In contrast, terminal plate **134A** may include passages adapted and located to only introduce a gas stream to the first anode of a DMEA **132A**, and terminal plate **134N+1** may include passages adapted and located to only remove a gas flow from a second cathode of a DMEA **132N**. In one aspect, terminal plate **134A** and/or terminal plate **134N+1** may not have passages, that is, they may be devoid of passages.

(60) According to aspects of the invention, as disclosed herein, the reduction reaction associated with the cathodes pursuant to Equation 2 may provide the electrons (e.sup.-) for the oxidation reaction associated with the anodes, for example, the prior or previous anode in the stack, pursuant to Equation 1. In one aspect, this flow of electrons (e.sup.-) in hydrogen purifier stack **130** is such that the DMEAs of hydrogen purifier stack **130** are electrically in series. However, in order to facilitate illustration and disclosure of this aspect of the invention, the flow of electrons (e.sup.-) from the second anodes to the first cathodes and the flow of electrons (e.sup.-) from the first anodes to the second cathodes, for example, of neighboring cells, in hydrogen purifier stack **130** are omitted from FIG. **4**.

(61) Hydrogen purifier stack **130** also includes opposing bus bars or current collectors **136** and **138** and end plates **137** and **139** (shown in phantom in FIG. **4**). Current collectors **136** and **138** may be similar in design, size, and construction to current collectors **88** and **90** shown and disclosed with respect to FIG. **3**. End plates **137** and **139**, as known in the art, may be relatively thick metal plates that function to assist in compressing stack **130**, for example, via a plurality of mechanical fasteners (not shown) extending between end plates **137** and **139**. For instance, end plates **137** and **139** may be associated with a plurality of threaded bolts having threaded nuts which, upon tightening on the bolts, compress the stack **130** between plates **137** and **139**.

(62) As shown in FIG. **4**, a plurality of input or first gas streams **140A-140N** having a hydrogen gas (H.sub.2) content and non-hydrogen gas content may be introduced to DMEAs **132A-132N** via inlets and flow distribution passages in plates **134A-134N**. As shown in FIG. **4**, first gas streams **140A-140N** may be provided, for example, via one or more gas supply manifolds **141** (shown in phantom in FIG. **4**) for example, a common gas supply manifold. As also shown in FIG. **4**, a plurality of discharge or third gas streams **142A-142N** may be removed from DMEAs **132A-132N**, for example, via flow extraction passages and outlets in plates **134A-134N**. As shown in FIG. **4**,

third gas streams **142A-142N** may be discharged, for example, via one or more gas collection manifolds **143** (shown in phantom in FIG. **4**), for example, a common gas collection manifold. As disclosed herein, the plurality of output or third gas streams **142A-142N** have a higher hydrogen gas (H.sub.2) content and a lower non-hydrogen gas content than the plurality of input gas streams **140A-140N**, respectively. For example, as disclosed herein, the non-hydrogen gas content of third gas streams **142A-N** may have at most 100 ppm of non-hydrogen gas, or at most 20 ppm, or at most 10 ppm, or at most 5 ppm, or at most 2 ppm, or at most 1 ppm, or even at most 500 ppb, “on a dry basis.” or even at most 500 ppb, “on a dry basis.” According to aspects of the invention, each of the flow of gas streams from the input flows **140A-140N** to the output flows **142A-142N** may flow in parallel through the DMEAs **132A-132N**, for example, from the one or more gas supply manifolds **141** to the one or more gas collection manifolds **143**.

(63) Though not shown in FIG. **4** to facilitate illustration and disclosure of the invention, according to aspects of the invention, hydrogen purifier stack **130** may typically include a plurality of anode exhaust ports, passages, and/or manifolds to allow excess source gas from streams **140A-140N** to be allowed to be discharged from hydrogen purifier stack **130**.

(64) As also shown in FIG. **4**, hydrogen purifier stack **130** may be powered by a voltage, ΔV , and a current, I , between current collectors **136** and **138**. The voltage ΔV may be provided externally, for example, from one or more DC power supplies, the local electrical grid, a fuel cell, by photovoltaics, and/or wind turbine; and/or internally, for example, from electro-chemical potentials and reactions occurring in DMEAs **132A-132N** in stack **130**. In one aspect, the voltage ΔV may be regulated to regulate the current through stack **130**. A higher electric current, may enhance the chemical reactions within stack **130** and increase the hydrogen gas production rate. The amount of amperage required to enhance the output of aspects of the invention will be dependent, among other things, upon the size of the purifier and the number of cells in the purifier.

(65) As also shown in FIG. **4**, according to one aspect of the invention, one or more DMEAs **132A-132N** in hydrogen purifier stack **130** may comprise a purifier cell **50** shown and described with respect to FIG. **2**. That is, in one aspect, one or more the DMEAs **132A-132N**, for example, all of DMEAs **132A-132N**, may be adapted to remove at least some gas stream, for example, the second gas stream **38** shown in FIG. **2**, evolved from the first cathode of DMEAs **132A-132N**. This removal of at least some of the gas stream from the second gas stream of DMEAs **132A-132N** is indicated by gas streams **144A-144N** (shown in phantom) in FIG. **4**. Gas streams **144A-144N** in FIG. **4** correspond to gas flow stream **58** in FIG. **2**. As disclosed herein, according to an aspect of the invention, it is understood that the removal of gas stream **144A-144N**, having at least some non-hydrogen gas content, reduces the partial pressure of the non-hydrogen gas introduced to a second anode of the DMEAs **132A-132N** to thus reduce the passage of the non-hydrogen gas through the second anode. According to one aspect, gas flow streams **144A-144N** may be discharged from hydrogen purifier stack **130** via discharge passages in hydrogen purifier stack **130**, for example, a discharge gas manifold (not shown in FIG. **4**.)

(66) In addition, since some of the gas flow streams **144A-144N** (in phantom) in FIG. **4** may contain some hydrogen gas content, in one aspect, at least some hydrogen gas may be introduced to the second gas streams (that is, second gas stream **38** shown in FIG. **2**) in DMEAs **132A-132N** to replenish at least some of the lost hydrogen gas and enhance the content of the hydrogen gas introduced to the second anodes of DMEAs **132A-132N**. By replenishing any hydrogen gas lost via gas flow streams **144A-144N**, aspects of the present invention can enhance the hydrogen content of the hydrogen gas produced by hydrogen purifier stack **130**. This introduction of at least some hydrogen gas to the second gas streams of DMEAs **132A-132N** is indicated by gas streams **146A-146N** (in phantom) in FIG. **4**. According to one aspect, gas streams **146A-146N** may be introduced to hydrogen purifier stack **130** via passages in hydrogen purifier stack **130**, for example, a gas manifold (not shown in FIG. **4**.)

(67) FIG. **5** is a schematic illustration of a hydrogen gas purifier cell **150** according to a further

aspect of the invention. According to this aspect, purifier cell **150** may have many of the features of purifier cell **10** and purifier cell **50** disclosed herein. Specifically, purifier cell **150** includes an DMEA **152** positioned to receive an input or first gas stream **154** having a first hydrogen gas content **156** and a first non-hydrogen gas content **158** and yields an output or third gas stream **160** having an output or third hydrogen gas content **162** greater than the first hydrogen gas content **156** and an output or third non-hydrogen gas content **164** less than the first non-hydrogen gas content **158**. As is typical in the art, at least some of input or first gas stream **154** may not oxidize at anode **166**, but be removed as gas stream **155**, for example, an “exhaust gas stream.” Typically, exhaust stream **155** may be captured and directed, for example, via channels, manifolds, and ports, to further processing or disposed of as needed.

(68) Again, output or third gas stream **160** will typically have a greater pressure than the pressure of input or first gas stream **154**, though the pressure may be lower than the pressure of the first gas stream **154**. However, according to this aspect of the invention, DMEA **152** of purifier cell **150** includes only three electrodes, where the second electrode performs the dual function of both a first cathode and a second anode, as disclosed herein.

(69) Specifically, as shown in FIG. 5, DMEA **152** of purifier cell **150** includes a first anode **166**, a first electrolyte **168**, a first cathode/second anode (or “dual electrode”) **170**, a second electrolyte **172** and a second cathode **174**. In a fashion similar to the components and operation of cell **10** and cell **50** disclosed herein, first anode **166** may have all the features of the first anodes disclosed herein; first electrolyte **168** and second electrolyte **172** may have all the features of the electrolytes disclosed herein; and second cathode **174** may have all the features of the second cathodes disclosed herein.

(70) In addition, dual electrode **170** of cell **150** may have all the features and characteristics of an anode or a cathode disclosed herein, for example, being electrically conductive, gas-permeable, and containing a catalyst. However, according to this aspect of the invention, dual electrode **170** performs the dual function of first reducing the hydrogen ions ($H_{sup.}^{+}$) transferred through first electrolyte **168** with electrons ($e_{sup.}^{-}$) to yield hydrogen gas ($H_{sub.2}$), according to Equation 2, and then oxidizing the hydrogen gas ($H_{sub.2}$) to yield hydrogen ions ($H_{sup.}^{+}$) and electrons ($e_{sup.}^{-}$), according to Equation 1. In one aspect, the electrons ($e_{sup.}^{-}$) produced in the oxidation of hydrogen gas ($H_{sub.2}$) in dual electrode **170** may be the source of electrons ($e_{sup.}^{-}$) consumed in the reduction of the hydrogen ions ($H_{sup.}^{+}$) in dual electrode **170**. This internal transfer of electrons ($e_{sup.}^{-}$) within dual electrode **170** in DMEA **152** is represented by the electron loop **176** shown in phantom in FIG. 5. According to aspects of the invention, 1 or more cells **150** may be used for cells **132A-132N** of stack **130** shown in FIG. 4.

(71) According to one aspect of the invention, a discharge gas stream (gas stream **58** in FIG. 2) may be removed from and/or make-up gas stream (gas stream **59** and/or **59A** in FIG. 2) may be introduced to dual electrode **170** of cell **150**. As shown in FIG. 5, a gas stream **178** may be removed from dual electrode **170**, for example, through the gas-permeable structure of dual electrode **170**, to reduce the partial pressure of the non-hydrogen gas passed through first electrode **168**, as disclosed herein. Also, with or without the removal of gas stream **178**, a make-up, hydrogen ($H_{sub.2}$) gas-containing gas stream **180** (shown in phantom in FIG. 5) may be introduced to dual electrode **170**, for example, through the gas-permeable structure of dual electrode **170**, to replace or enhance the hydrogen gas ($H_{sub.2}$) content of the gas stream introduced to second electrolyte **172**, as disclosed herein. As in other aspects of the invention, a make-up, hydrogen ($H_{sub.2}$) gas-containing gas stream **180** may be introduced to dual electrode **170** by “back diffusion” of at least some of gas stream **160** back through second electrode **172** to dual electrode **170**.

(72) In one aspect, in order to enhance the distribution of the gas streams and/or the hydrogen ions ($H_{sup.}^{+}$) about the electrode surfaces, one or more gas diffusion layers (GDL), not shown in FIG. 5, and/or flow field insert, as described with respect to FIG. 3B, may be positioned within cell **150**. For example, cell **150** may include a GDL and/or flow field insert associated with first anode **166**,

dual electrode **170**, and/or second cathode **174**.

(73) As in other aspects of the invention, though having only three electrodes **166**, **170**, and **174**, a third hydrogen gas content **162** of third gas stream **160** of hydrogen purifier cell **150** is greater than the first hydrogen gas content **156**; and the third impurity gas content **164** is less than the first impurity gas content **158**. For example, the third gas stream **160** may have a purity, on a “dry basis,” in percent that is greater than the purity of the first gas stream **154-156**. For example, in one aspect, the purity of the third gas stream **160** may be at least 99.99 percent hydrogen, by volume, or at least 99.999 (five 9s) percent, by volume, or 99.9999 (six 9s) percent by volume. According to another aspect of the invention, the purity of the third gas stream **160** may be expressed on a basis of the third impurity gas content **164**. For example, in one aspect the third impurity gas content **164** may be at most 100 ppm of impurity gas. In other aspects of the invention, the third impurity gas content **164** produced may be at most 20 ppm, or at most 10 ppm, or at most 5 ppm, or at most 2 ppm, or at most 1 ppm. In other aspects of the invention, the third impurity gas content **164** of the hydrogen gas produced may be at most 750 ppb (that is, at most 0.750 ppm), or at most 500 ppb, or at most 200 ppb, or even at most 100 ppb. As known in the art, these impurity contents of the hydrogen gas produced are typically “on a dry basis.”

(74) In addition, according to aspects of the invention, the resulting gas stream, that is, third gas stream **160** in FIG. 5, having a higher hydrogen gas content **162** and lower impurity gas content **164**, may typically be provided at a higher pressure than the pressure of the feed or first gas stream **154**. For example, according to aspects of the invention, where first gas stream **154** may have a pressure of at most 1 psig, third gas stream **160** may have a pressure of at least 150 psig. In one aspect, the pressure of third gas stream **160** may be at least 120 psig, or at least 200 psig, or even at least 10,000 [10 thousand] psig. In other aspects of the invention, the pressure of third gas stream **160** may not be greater than the pressure of the feed or first gas stream **154**; in one aspect, the pressure of the third gas stream **160** may be less than the pressure of the first gas stream **154**.

(75) FIG. 6 is a schematic view of a hydrogen gas purifier system **200** having one or more hydrogen purifiers **202** having one or more hydrogen purifier stacks **130** shown in FIG. 4, according to an aspect of the invention. The one or more hydrogen purifier stack **130** may include any one or more of the hydrogen purifier cells disclosed herein, for example, one or more cells **10**, one or more cells **50**, and/or one or more cells **150**, or a combination thereof. The one or more hydrogen purifier stacks **130** may be contained by end plates **204** and **206**, for example, similar to end plates **137** and **139** shown in FIG. 4.

(76) As shown in FIG. 6, hydrogen gas purifier system **200** includes a source of hydrogen-containing gas **208** operatively connected to one or more hydrogen purifiers **202** via one or more conduits or pipes **210**. The source of hydrogen-containing gas **208** may be a storage tank, another gas purifier **202**, one or more fuel cells, or one of various industrial processes. As disclosed herein, the source of hydrogen-containing gas **208** typically contains at least some non-hydrogen or impurity gas, such as, nitrogen (N_{2}), argon (Ar), carbon dioxide (CO_{2}), carbon monoxide (CO), methane (CH_{4}), and/or oxygen (O_{2}). As shown in FIG. 6, the flow gas from the source of hydrogen-containing gas **208** may be regulated by one or more flow control valves **212**, for example, a manual valve or an automated valve controlled by an appropriate control system (not shown), and/or a pressure regulator **213**. In one aspect, the hydrogen-containing gas **208** may be introduced to system **200** under pressure, for example, by a gas pressurizing device (not shown), such as, one or more blowers, fans, or compressors. In another aspect, the hydrogen-containing gas **208** may be drawn into system **200** by a gas pressure reducing device or vacuum device (not shown), such as, a one or more blowers, fans, or compressors. For example, a source of vacuum may be operatively connected to one or more conduits or pipes **214**; to one or more discharge conduits **220**; and/or to one or more discharge conduits **221** in FIG. 6.

(77) After introduction of the hydrogen-containing gas from source **208** and appropriate oxidation and reduction, as disclosed herein, the higher purity hydrogen gas is discharged from the one or

more hydrogen purifier stacks **202** into one or more conduits or pipes **214** to storage or further processing **216**. For example, further processing **216** may be a dryer or desiccator to remove at least some water vapor, and/or to further purification, for example, to remove at least some trace impurities, if present. The purification process may comprise a pressure swing absorption (PSA) system, a temperature swing absorption (TSA), a “getter” gas purifier, or to another gas purifier system **200**. In one aspect, further processing **216** may include a liquefier, for example, a cryo-cooler. As shown in FIG. 6, the flow of gas from the one or more hydrogen purifier stacks **202** may be regulated by one or more flow control valves **218**, for example, a manual valve or an automated valve controlled by appropriate control system (not shown).

(78) As also indicated in FIG. 6, typically, the hydrogen-containing gas from source **208** may typically have a pressure, or first pressure, $P_{sub.1}$, and after passing through one or more hydrogen purifier stacks **202** the gas introduced to storage **216** may typically have a higher pressure, or third or output pressure, $P_{sub.2}$ greater than the first pressure $P_{sub.1}$, as disclosed herein. Pressure $P_{sub.2}$ may be any one of the output pressures disclosed herein. In one aspect, pressure $P_{sub.2}$ may comprise a pressure lower than pressure $P_{sub.1}$, for example, when the hydrogen-containing gas **208** is drawn into system **200**, for instance, by vacuum.

(79) As also indicated in FIG. 6, in one aspect, the one or more hydrogen purifier stacks **202** may include one or more purifier cells **50** (and disclosed and described with respect to FIG. 2) having discharge gas streams or displaced gas streams **58** and, possibly, hydrogen-containing gas stream **59**. As shown in FIG. 6, gas purifier system **200** may include one or more discharge conduits **220** for removing gas stream **58** (FIG. 2) and one or more discharge conduits **221** for removing anode exhaust gas (for example, anode gas stream **13** shown in FIG. 1) from the one or more hydrogen purifier stacks **202** and direct the discharged gas stream in conduit **220** and/or exhaust gas stream in conduit **221** to other processing, storage, or disposal **222**. The other processing of the gas streams in conduits **220** and **221** may include purification, for example, via another system **200**, among others; venting to atmosphere; combustion via a “flare”; or other processing. As shown in FIG. 6, the flow of discharged gas via conduit **220** from the one or more hydrogen purifier stacks **202** may be regulated by one or more flow control valves **224**, for example, a manual valve or an automated valve controlled by appropriate control system (not shown). In one aspect, at least some of the discharged gas stream (stream **58** in FIG. 2) may be introduced to purifier **200**, such as the purifier **200** shown in FIG. 6 or another purifier **200**, to recover and/or purify any hydrogen gas in the discharged gas, for example, in conduit **220**. For example, as shown in FIG. 6, at least some of the discharged gas (stream **58** in FIG. 2) in conduit **220** may be forwarded to input conduit **210** via conduit or pipe **223** (shown in phantom in FIG. 6). The flow in conduit **223** may include appropriate flow control and/or pressure control (not shown) as disclosed herein.

(80) In one aspect, the one or more purifier cells **50** may be provided with hydrogen-containing “make-up” gas stream **59** (for example, see FIG. 2) from source **226**, such as, from a discharge conduit **214** from this or another system **200** or another hydrogen gas source, via one or more conduits or pipes **228**. The flow of “make up” gas through one or more conduits **228** to the one or more hydrogen purifier stacks **202** may be regulated by one or more flow control valves **230**, for example, a manual valve or an automated valve controlled by appropriate control system (not shown). In one aspect, the flow of make-up gas (stream **59** in FIG. 2) may not be sourced externally, as indicated in FIG. 6, but may also be sourced and routed internally in stack **130**, for example, from one or more cells **50** (see FIG. 2) to one or more cells **50**, the flow of which may be regulated and controlled based upon flow and/or pressure.

(81) According to one aspect of the invention, the temperature of the one or more one or more hydrogen purifier stacks **130** of gas purifier system **200** may be regulated and controlled to, among other things, optimize the purification performance of the DMEAs of the stacks **130** and/or avoid overheating of the stacks **130**. For example, in one aspect, the temperature of the stacks **130** may be maintained at at least 30 degrees C., but is typically maintained at range of between 50 degrees C.

to 80 degrees C. It is understood that reducing the operating temperature of the electrolytes in the cells of stacks **130**, for example, to between 45 degrees C. to 55 degrees C., may enhance the purity of the hydrogen gas produced, while requiring a more reasonable electrical power consumption. In one aspect, as shown in FIG. **6**, the temperature of the one or more hydrogen purifier stacks **130** may be monitored and regulated by one or more temperature sensors **232**. The temperature of the one or more hydrogen purifier stacks **130** detected by temperature sensor **232** may be controlled by one or more heating or cooling circuits regulated and controlled by an appropriate control system (not shown). The heating or cooling circuits of the one or more hydrogen purifier stacks **130** may include passages in the one or more hydrogen purifier stacks **130** through which heating or cooling fluid may be passed to regulate the temperature of the one or more hydrogen purifier stacks **130**.

(82) According to one aspect of the invention, the pressure of the one or more hydrogen purifier stacks **130**, the pressure of the gas from the source **208**, and/or pressure of the discharge gas of gas purifier system **200** may be regulated and controlled to, among other things, optimize the performance of the reactions within the DMEAs of the stacks **130**. For example, one aspect, the pressure of the hydrogen gas produced by the one or more stacks **130** may be monitored and regulated by one or more pressure sensors **234**. It is understood that in some aspects the pressure of hydrogen gas generated in system **200** may affect the purity of the hydrogen gas produced. The pressure of the hydrogen gas produced that is detected by pressure sensor **234** may be regulated and controlled by an appropriate control system (not shown).

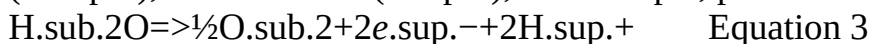
(83) In addition, in one aspect, the pressure of the one or more stacks **130** may be regulated and controlled to enhance the performance, for example, the efficiency, of the DMEAs of the stacks **130**. In one aspect, the pressure of the first cathode and/or the second anode of the one or more hydrogen purifier stacks **130** may be regulated and controlled to enhance the performance, for example, the efficiency, of the DMEAs of the stacks **130**. In one aspect, as shown in FIG. **6**, the pressure of one or more stacks **130**, for example, the pressure of the first cathode and/or the second anode, may be monitored and regulated by one or more pressure sensors **236**. The pressure of the one or more hydrogen purifier stacks **130** detected by pressure sensor **236** may be controlled by an appropriate control system (not shown).

(84) FIG. **7** is a schematic illustration of a water electrolyzer cell **250** according to another aspect of the invention. According to this aspect, a water electrolyzer cell **250** cell comprises a electrochemical cell for electrolyzing water to generate hydrogen gas, specifically, a hydrogen gas having less undesirable impurity gas, such as, oxygen (O.sub.2). As shown in FIG. **7**, water electrolyzer cell **250** is positioned and adapted to receive a feed or first fluid stream **252** containing water. First fluid stream **252** may contain liquid water and/or gaseous water (that is, steam). (It is to be understood that any reference to “fluid” herein may refer to a liquid fluid, a gaseous fluid, or both a liquid fluid and a gaseous fluid.) According to aspects of this invention, cell **250** is adapted to produce a hydrogen gas stream or third gas stream **254** having little or no gas impurities, including little or no oxygen gas. The third gas stream **254** may be referred to as a discharge gas stream **254**.

(85) As shown schematically in FIG. **7**, water electrolyzer cell **250**, in a fashion similar to the hydrogen gas purifier cells disclosed herein, typically comprises a multilayer structure having components, for example, anodes and cathodes having thin planar or thin laminar construction, where the structures shown in FIG. **7** may comprise a side elevation view or a trans-axial cross-sectional view of water electrolyzer cell **250** that is not drawn to scale, but is drawn to facilitate disclosure of the invention.

(86) According to aspects of the invention, to provide the desired hydrogen gas **254**, water electrolyzer cell **250** typically includes a first MEA **256** and at least one second MEA **258**. The first MEA **256** includes a first anode **260**, which, in a fashion similar to other anodes disclosed herein, may typically be fluid-permeable, specifically, water- or gas-permeable, where at least some of the water in first fluid stream **252** may pass into anode **260**, for example, in an axial direction as

indicated by the arrow of first fluid stream 252. In addition, anode 260 includes at least some catalyst, for example, at least some platinum group metal-containing or iridium-containing catalyst, capable of enhancing the oxidation of the water to gaseous oxygen (O.sub.2), hydrogen ions (H.sup.+), and electrons (e.sup.-), for example, pursuant to equation 3.



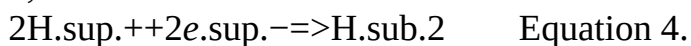
(87) In one aspect, in order to enhance the distribution of first fluid stream 252 about the surface of anode 260, an electrically-conductive gas diffusion layer (GDL) and/or flow field insert, not shown in FIG. 7, may be positioned between first fluid stream 252 and anode 260, for example, the GDL may be applied over the surface of anode 260 contacted by first fluid stream 252. In one aspect, the GDL used for the cell 250, or the GDL layer used in any aspect disclosed herein, may be a metal-based GDL, for example, a platinum-coated titanium GDL, or its equivalent.

(88) Due to the permeability of anode 260, water (H.sub.2O) passes into anode 260, and due to the electrical conductivity of anode 260, electrons (e.sup.-) are conducted away from anode 260, and according to aspects of the invention, the hydrogen ions (H.sup.+) generated in anode 260 are introduced to an electrolyte 262. Electrolyte 262 may be similar to any of the other electrolyte disclosed herein, for example, electrolyte 262 may be an acidic electrolyte.

(89) As is typical in the art, at least some of input or first fluid stream 252 may not diffuse through anode 260, but be removed as fluid stream 264, for example, an “exhaust gas stream.” Typically, exhaust stream 264 may be captured and directed, for example, via channels, manifolds, and ports, to further processing or disposed of as needed.

(90) Electrolyte 262, or a first electrolyte 262, is positioned and adapted to receive and transfer at least some of the hydrogen ions (H.sup.+). Due to the close proximity of anode 260 to electrolyte 262, hydrogen ions (H.sup.+) are passed from anode 260 to electrolyte 262. First electrolyte 262 comprises a barrier between the first anode 260 and the electrode 266. First electrolyte 262 may comprise any material or substance capable of transmitting the hydrogen ions (H.sup.+), for example, selectively transmitting hydrogen ions (H.sup.+), that is, protons, from first anode 260 to electrode 266. That is, in one aspect, electrolyte 262, and any electrolyte disclosed herein, may be referred to as a “proton-conductive material,” while substantially preventing the flow of gas. First electrolyte 262 may typically be acidic, for example, an acidic polymer containing a perfluorosulfonic acid (PFSA). In one aspect, electrolyte 262 may be a membrane marketed under the trademark Nafion™ by The Chemours Company of Wilmington, Delaware, or its equivalent. In other aspects, electrolyte 262 may contain one or more of the following acids: phosphoric acid [H.sub.3PO.sub.4], sulfuric acid [H.sub.2SO.sub.4], or any other hydrogen ion (H.sup.+) conducting acid. In one aspect, first electrolyte 262 may comprise a proton exchange membrane (PEM), as known in the art.

(91) Electrode 266, which may be referred to as “cathode” 266, as known in the art, or a first cathode 266, is positioned to receive at least some of the hydrogen ions (H.sup.+) transferred by the first electrolyte 262. Similar to other cathodes disclosed herein, first cathode 266 is fluid permeable and typically contains a catalyst, for example, a platinum group-containing catalyst, adapted to enhance the reaction (that is, reduction) of at least some of the hydrogen ions (H.sup.+) that pass through first electrolyte 262 by reacting with at least some electrons (e.sup.-) pursuant to Equation 4, as known in the art.



(92) The resulting, or “evolved,” hydrogen gas (H.sub.2) 268, or a second fluid stream 268 containing hydrogen gas (H.sub.2), is transferred through gas permeable first cathode 266 as indicated as gas stream 268 in FIG. 7, or a second fluid stream 268.

(93) In one aspect, in order to enhance the distribution of the hydrogen gas (H.sub.2) about the surface of second anode 270, a gas diffusion layer (GDL) and/or flow field insert, not shown in FIG. 7, may be positioned between first cathode 266 and second anode 270, for example, the GDL and/or flow field insert may be applied over the surface of second anode 270.

(94) As shown in FIG. 7, according to aspects of the invention, second fluid stream **268** containing hydrogen gas (H.sub.2) may then be introduced to second MEA **258**, specifically, to electrode **270**, or second anode **270**, of second MEA **258**. In the schematic diagram of the electrolyzer cell **250** shown in FIG. 7, first MEA **256** is shown spaced from second MEA **258** to facilitate illustration and disclosure of the invention. However, according to aspects of the invention, the spacing between first MEA **256** and second MEA **270** may be minimal, for example, where the surface of first cathode **266** may abut or contact the surface of second anode **270**. However, in one aspect, at least some spacing, for example, 0.1 millimeters [mm] to 0.5 mm, may be present between the surface of first cathode **266** and the surface of second anode **270**.

(95) According to aspects of the invention, and contrary to the existing art, second fluid stream **268** having hydrogen gas (H.sub.2) may then be introduced to second MEA **258** without removing or extracting second fluid stream **268** from electrolyzer cell **250**. In other words, though some portion of second fluid stream **268** may undesirably “escape” from electrolyzer cell **250**, according to one aspect of the invention, substantially all of second fluid stream **268** produced at first cathode **266** is received by second anode **270**. For example, in one aspect, second fluid stream **268** may be allowed to pass from first cathode **266** of MEA **256** to second anode **270** of MEA **258** without any intervening handling or treatment, for instance, without passing outside of electrolyzer cell **250**. In one aspect, the evolved hydrogen gas (H.sub.2) formed in first cathode **266** may substantially be immediately oxidized to hydrogen ions (H.sup.+) at second anode **270**. In one aspect, second fluid stream **268** may be allowed to pass directly from first cathode **266** of MEA **258** to second anode **270** of MEA **258** of electrolyzer cell **250**, for instance, without passing outside of electrolyzer cell **250** before reaching second cathode **270**.

(96) The second anode **270** is positioned to be contacted by the second fluid stream **268** having hydrogen gas (H.sub.2). Second anode **270** of second MEA **258** may be similar to, if not identical to, first anode **260** of MEA **256**. Second anode **270** may be hydrogen-gas permeable, where at least some of the hydrogen gas in second fluid stream **268** may pass through second anode **40**, for example, in an axial direction, as indicated by the arrow of second fluid stream **268**. In addition, second anode **270** includes at least some catalyst, for example, at least some platinum-containing catalyst, capable of enhancing the oxidation of hydrogen gas to hydrogen ions (H.sup.+), as indicated by Equation 1, which is reproduced below.

$$\text{H.sub.2} \Rightarrow 2\text{H.sup.} + 2\text{e.sup.-}$$
 Equation 1.

(97) In one aspect, in order to enhance the distribution of second fluid stream **268** about the surface of second anode **270**, a gas diffusion layer or GDL and/or flow field insert, not shown in FIG. 7, may be positioned between second fluid stream **268** and second anode **270**, for example, the GDL and/or flow field insert may be applied over the surface of second anode **270** contacted by second fluid stream **268**.

(98) According to aspects of the invention, the catalyst contained in second anode **270** promotes or enhances the oxidation of the hydrogen gas in second fluid stream **268** to yield or produce hydrogen ions (H.sup.+) and electrons (e.sup.-) pursuant to Equation 1, above. In one aspect, the electrons (e.sup.-) generated in second anode **270** may be directed back to first cathode **266**, as indicated by arrow **272** in FIG. 7, to supply at least some of the electrons (e.sup.-) used to reduce the hydrogen ions (H.sup.+) to hydrogen (H.sub.2) at first cathode **266**, pursuant to Equation 4. Due to the permeability of second anode **270**, at least some of the hydrogen gas (H.sub.2) generated in second anode **270** may pass through second anode **270**, and, according to aspects of the invention, is introduced to or come in contact with an electrolyte **274**.

(99) Electrolyte **274** of MEA **258**, or a second electrolyte **274**, is positioned and adapted to receive and transfer at least some of the hydrogen ions (H.sup.+) received from the second anode **270**. Second electrolyte **274** comprises a gas barrier between the second anode **270** and the cathode **276**. Second electrolyte **274** may be similar to, if not substantially identical to, first electrolyte **262**, and comprise any material or substance capable of transmitting the hydrogen ions (H.sup.+), for

example, selectively transmitting hydrogen ions (H.sup.+), that is, protons, from second anode **270** to an cathode **276**. Again, as noted with respect to first electrolyte **262**, in one aspect, second electrolyte **274** may be referred to as a “proton-conductive material.” Second electrolyte **274** may typically be acidic, for example, containing one or more of the acids identified herein. However, in one aspect, second electrolyte **274** may comprise a PEM, as known in the art.

(100) Electrode **276**, or second cathode **276**, is positioned to receive at least some of the hydrogen ions (H.sup.+) transferred by second electrolyte **274**. Similar to the second anode **266**, second cathode **276** is typically permeable and contains a catalyst, for example, a platinum-containing catalyst, adapted to enhance the reduction of at least some of the hydrogen ions (H.sup.+) using at least some electrons (e.sup.-) pursuant to Equation 2, which is reproduced below.

$2\text{H.sub.2} + 2\text{e.sup.-} \Rightarrow \text{H.sub.2}$ Equation 2.

The resulting, or “evolved,” hydrogen gas (H.sub.2), or a third fluid stream **254** containing hydrogen a gas (H.sub.2), is produced by electrolysis cell **250**.

(101) In one aspect, in order to enhance the distribution of the hydrogen ions (H.sup.+) about the surface of second cathode **276**, a gas diffusion layer (GDL) or a flow field insert, not shown in FIG. 7, may be positioned between second electrolyte **274** and second cathode **276**, for example, the GDL and/or flow field insert may be applied over the surface of second cathode **276**. In one aspect, a GDL or a flow field insert, not shown in FIG. 7, may be positioned adjacent to the downstream surface of second cathode **276** to enhance the removal of third fluid stream **254**.

(102) According to aspects of the invention, the third fluid stream **254** of electrolysis cell **250** may typically contain hydrogen gas (H.sub.2) with little or no undesirable impurity gases, for example, little or no oxygen gas.

(103) According to aspects of the invention, in electrolysis cell **250**, bypassing the second fluid stream **268** from first anode **266** of MEA **256** to second cathode **270** of second MEA **258** can provide a more efficient device providing a purer hydrogen gas than the prior art. Among other things, the passing of the second fluid stream **268** from first cathode **266** to second anode **270**, for example, directly, minimizes the content of impurities, for example, oxygen gas impurities, in the third fluid stream **254**.

(104) FIG. 8 is a schematic illustration of a water electrolyzer cell **300** according to another aspect of the invention. According to this aspect, electrolyzer cell **300** may have many of the features of electrolyzer cell **250**; however, electrolyzer cell **300** further includes at least one gas discharge or release between the MEAs. It is believed that this reduction in the second fluid stream content reduces the partial pressure of the non-hydrogen gas, for example, the partial pressure of oxygen, between the MEAs and thus reduces the partial pressure gradient driving force of the undesirable non-hydrogen gas through the second MEA. In a further aspect, since some of the desirable hydrogen gas will be lost when removing some of the second fluid stream, at least some hydrogen gas may be introduced between the MEAs to function as a “make-up” hydrogen gas for the hydrogen gas possibly lost with the removal of some of the second fluid stream.

(105) As shown in FIG. 8, water electrolyzer cell **300** is positioned and adapted to receive a feed or first fluid stream electrolyzer cell **302**. According to aspects of this invention, cell **300** is adapted to produce a hydrogen gas stream or third fluid stream **304** having little no gas impurities, including little or no oxygen gas. The third gas stream **254** may be referred to as a discharge gas stream **254**. As shown in FIG. 8, similar to electrolyzer cell **250**, typically comprises a multilayer structure having components, for example, anodes and cathodes having thin planar or thin laminar construction, where the structures shown in FIG. 8 may comprise a side elevation view or a trans-axial cross-sectional view of electrolyzer cell **300** that is not drawn to scale, but is drawn to facilitate disclosure of the invention.

(106) In one aspect, electrolyzer cell **300** shown in FIG. 8 may have substantially the same first MEA **306** as in electrolyzer cell **250**, that is, having a first anode **308**, a first electrolyte **310**, and a first cathode **312** to oxidize and reduce first fluid stream **302** to yield a second fluid stream **314**

containing hydrogen gas (H.sub.2) in a manner substantially identical to MEA 256 of electrolyzer cell 250 shown in FIG. 7. As is typical in the art, at least some of input or first fluid stream 302 may not oxidize at first anode 308, but may be removed as fluid stream 303, for example, an "exhaust fluid stream." In addition, electrolyzer cell 300 may have substantially the same second MEA 316 having a second anode 318, a second electrolyte 320, and a second cathode 322 to oxidize and reduce second fluid stream 314 to yield third fluid stream 304 containing hydrogen gas (H.sub.2) in a manner substantially identical to electrolyzer cell 250 shown in FIG. 7. However, according to the aspect of the invention shown in FIG. 8 electrolyzer cell 300 includes at least one discharge gas stream or displaced gas stream 324 having hydrogen gas and non-hydrogen gas, such as, oxygen.

(107) In one aspect, in order to enhance the distribution of the gas streams about the electrode surfaces, one or more gas diffusion layers (GDL) and/or gas-distribution media, not shown in FIG. 8, may be positioned within electrolyzer cell 300. For example, electrolyzer cell 300 may include a GDL and/or flow field insert associated with first anode 308, first cathode 312, second anode 318, and/or second cathode 322.

(108) According to this aspect, after first fluid stream 302 is processed and passed through first MEA 306 to produce second fluid stream 314 having a hydrogen gas content and a non-hydrogen gas content 36, at least some of the second fluid stream 314 is removed via fluid stream 324. The removal of fluid stream 324 from fluid stream 314 yields a modified or intermediate fluid stream 326. According to aspects of the invention, removal of fluid stream 324 reduces the partial pressure of non-hydrogen gas in modified fluid stream 326 and this reduction in partial pressure reduces the partial pressure gradient across the second MEA 316 of the non-hydrogen gas content, which reduces the passage of the non-hydrogen gas through second MEA 316 to third fluid stream 304.

Accordingly, according to aspects of the invention, the non-hydrogen gas content, for example, the oxygen gas content, of third fluid stream 304 is reduced providing a purer stream of hydrogen gas.

(109) The removal of fluid stream 324 may be practiced by various means. In one aspect, fluid stream 324 may be removed by simply venting at least some of second fluid stream 314, for example, venting through the inherent spacing between first MEA 306 and second MEA 316, for instance, via the inherent spaces or voids between the surface of first cathode 312 and second anode 318. In another aspect, fluid stream 324 may be removed by providing paths, channels, or grooves, for example, radial or transverse channels or grooves, in the mating surface of first cathode 312, in the mating surface of second anode 318, or in both the mating surface of first cathode 312 and in the mating surface of second anode 318. In another aspect, fluid stream 324 may be removed through a GDL and/or flow field insert positioned between the mating surface of first cathode 312 and the mating surface of second anode 318. As known in the art, a GDL is typically a porous material, such as, carbon paper, through which gas stream 324 may pass. In another aspect, fluid stream 324 may be removed by providing one or more spacers between the mating surface of first cathode 312 and the mating surface of second anode 318 to provide a pathway for fluid stream 324. In one aspect, a source of sub-atmospheric pressure, that is, vacuum, may be introduced to draw out at least some of the second fluid stream 314 via fluid stream 324.

(110) According to another aspect of the invention, after first fluid stream 302 is processed and passed through first MEA 306 to produce second fluid stream 314 at least some hydrogen gas may be introduced to second fluid stream 314. As shown in FIG. 8, in one aspect, hydrogen gas may be introduced to second fluid stream 314 via fluid stream 328 (shown in phantom in FIG. 8) to replace the hydrogen gas lost from second fluid stream 314 via fluid stream 324 and produce modified fluid stream 326. Though in one aspect, fluid stream 328 may be highly pure hydrogen gas, for example, having a purity to at least greater than the hydrogen content of first fluid stream 302; in other aspects, fluid stream 328 may be a hydrogen-gas containing stream having at least some hydrogen gas content, but may have a non-hydrogen gas content.

(111) The introduction of hydrogen-containing gas stream 328 to electrolysis cell 300, for example,

a “make-up gas stream,” may be practiced with or without the removal of fluid stream **324**. The introduction of fluid stream **328** to second fluid stream **314** may be practiced in any one or more convenient manners, for example, by introducing the hydrogen-gas containing stream **328** through electrolyte **310**, which may be gas-permeable, for example, driven by a hydrogen gas partial pressure gradient, or through the gas-permeable first cathode **312**, through a gas permeable GDL and/or flow field insert, or through channels in first cathode **312**, channels in second anode **318** or through channels in both first cathode **312** and in second anode **318**. Any channels that may be provided for hydrogen-gas containing stream **328** may be located in one or both of the opposing surfaces of first cathode **312** and second anode **318**.

(112) In one aspect, the make-up hydrogen gas stream **328** may comprise at least some of the third fluid stream **304** having a hydrogen content. For example, at least some of the third fluid stream **304** may be introduced to the second fluid stream **314** by diffusion through the second electrolyte **320**, as indicated in phantom by gas stream **328A** in FIG. **8**. This diffusion through the second electrolyte **320** may be referred to as “back diffusion” of at least some of the third fluid stream **304** having at least some hydrogen gas (H.sub.2). The make-up gas stream **328** may be provided by any one or more of these sources or mechanisms.

(113) In another aspect of the invention, the three-electrode cell **150** shown in FIG. **5** may also function as an electrolyzer. For example, as shown in FIG. **5**, DMEA **152** of purifier cell **150** includes a first anode **166**, a first electrolyte **168**, a first cathode/second anode (or “dual electrode”) **170**, a second electrolyte **172** and a second cathode **174**. In a fashion similar to the components and operation of electrolyzer cell **250** shown in FIG. **7**, in electrolyzer cell **150** shown in FIG. **5**, anode **166** may have all the features of the first anodes disclosed herein; first electrolyte **168** and second electrolyte **172** may have all the features of the electrolytes disclosed herein; and second cathode **174** may have all the features of the second cathodes disclosed herein. In this aspect of the invention, the components and fluid flows of electrolyzer cell **150** shown in FIG. **5** may have all the functions and characteristics of the purifier cell **150** disclosed herein, including the oxidation at first anode **166** pursuant in Equation 3 and the reduction at the second cathode **174** pursuant to Equation 4. However, for electrolyzer cell **150**, first fluid stream **154** in FIG. **5** may contain liquid water and/or gaseous water (that is, steam). (It is to be understood that any reference to “fluid” herein may refer to a liquid fluid, a gaseous fluid, or both a liquid fluid and a gaseous fluid.) According to aspects of this invention, electrolyzer cell **150** of FIG. **5** may be adapted to produce a hydrogen gas stream or third fluid stream **160** having little or no gas impurities, including little or no oxygen gas.

(114) According to aspects of the invention, one or more water electrolysis cells **250** and **300** may be provided to generate high purity hydrogen gas. In one aspect, a water electrolyze stack having one or more electrolysis cells **250** and/or **300** may be provided, for example, a water electrolysis stack similar to hydrogen purifier stack **130** shown in FIG. **4**. Also, in one aspect, a water electrolyzer system having one or more electrolysis cell stacks having one or more electrolysis cells **250** and/or **300** may be provided, for example, a water electrolysis similar to hydrogen purifier system **200** shown in FIG. **6**.

(115) As disclosed above, embodiments of the present invention, in their many aspects, provide improved hydrogen purification and water electrolysis that meets, and can exceed, the hydrogen gas purities required by hydrogen gas users. Aspects of the present invention employ a unique combination of membrane electrode assemblies (MEAs) or double MEAs (DMEAs) that have shown to provide the enhanced hydrogen gas purities needed.

(116) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements,

and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(117) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

(118) While several aspects of the present invention have been described and depicted herein, alternative aspects may be effected by those skilled in the art to accomplish the same objectives. Accordingly, it is intended by the appended claims to cover all such alternative aspects as fall within the true spirit and scope of the invention.

Claims

1. A hydrogen gas purifier cell comprising: a first membrane electrode assembly (MEA) comprising: a first anode positioned to contact a first gas stream having a first hydrogen gas content and a first impurity gas content, the first anode containing a catalyst adapted to oxidize at least some of first hydrogen gas content to produce hydrogen ions and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the first anode; and a first cathode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the first cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content greater than the first hydrogen gas content and a second impurity gas content less than the first impurity gas content; a second MEA comprising: a second anode positioned adjacent the first cathode to receive the second gas stream from the first cathode of the first MEA, the second anode containing a catalyst adapted to oxidize at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the second anode; and a second cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte of the second MEA, the second cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content.
2. The hydrogen gas purifier cell as recited in claim 1, wherein the purifier cell further comprises at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second gas stream.
3. The hydrogen gas purifier cell as recited in claim 2, wherein the at least one passage is located between the first cathode and the second anode.
4. The hydrogen gas purifier cell as recited in claim 3, wherein the at least one passage located between the first cathode and the second anode comprises voids between mating surfaces of the first cathode and the second anode.
5. The hydrogen gas purifier cell as recited in claim 3, wherein the purifier cell further comprises a gas-permeable layer (GDL) between the first cathode and the second anode, and wherein the GDL provides the at least one passage located between the first cathode and the second anode.
6. The hydrogen gas purifier cell as recited in claim 5, wherein the GDL comprises a carbon-based, gas-permeable layer.

7. The hydrogen gas purifier cell as recited in claim 2, wherein the at least one passage comprises at least one channel in close proximity to at least one of the first cathode and the second anode.
8. The hydrogen gas purifier cell as recited in claim 1, wherein the purifier cell further comprises at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream.
9. The hydrogen gas purifier cell as recited in claim 8, wherein the at least one passage is located between the first cathode and the second anode.
10. The hydrogen gas purifier cell as recited in claim 9, wherein the at least one passage located between the first cathode and the second anode comprises voids between mating surfaces of the first cathode and the second anode.
11. The hydrogen gas purifier cell as recited in claim 1, wherein the second gas stream passes directly from the first cathode to the second anode.
12. The hydrogen gas purifier cell as recited in claim 11, wherein the second gas stream passes directly from the first cathode to the second anode without passing outside of the hydrogen gas purifier cell.
13. The hydrogen gas purifier cell as recited in claim 1, wherein the spacing between the second anode and the first cathode is minimal.
14. The hydrogen gas purifier cell as recited in claim 1, wherein the spacing between the second anode and the first cathode is 0.1 mm to 0.5 mm.
15. The hydrogen gas purifier cell as recited in claim 1, wherein the second anode contacts the first cathode.
16. The hydrogen gas purifier cell as recited in claim 1, wherein the second gas stream passes from the first cathode to the second anode without any intervening handling.
17. A hydrogen gas purifier cell comprising: a first membrane electrode assembly (MEA) comprising: a first anode positioned to contact a first gas stream having a first hydrogen gas content and a first impurity gas content, the first anode containing a catalyst adapted to oxidize at least some of first hydrogen gas content to produce hydrogen ions and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the first anode; and a first cathode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the first cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content greater than the first hydrogen gas content and a second impurity gas content less than the first impurity gas content; a second MEA comprising: a second anode positioned to receive the second gas stream from the first cathode of the first MEA, the second anode containing a catalyst adapted to oxidize at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the second anode; and a second cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte of the second MEA, the second cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content; and at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second gas stream, the at least one passage comprising voids between mating surfaces of the first cathode and the second anode.
18. The hydrogen gas purifier cell as recited in claim 11, wherein the purifier cell further comprises at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream.
19. The hydrogen gas purifier cell as recited in claim 18, wherein the at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream comprises the voids between the mating surfaces of the first cathode and the

second anode.

20. A hydrogen gas purifier cell comprising: a first membrane electrode assembly (MEA) comprising: a first anode positioned to contact a first gas stream having a first hydrogen gas content and a first impurity gas content, the first anode containing a catalyst adapted to oxidize at least some of first hydrogen gas content to produce hydrogen ions and electrons; a first electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the first anode; and a first cathode positioned to receive at least some of the hydrogen ions transferred by the first electrolyte, the first cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a second gas stream having a second hydrogen gas content greater than the first hydrogen gas content and a second impurity gas content less than the first impurity gas content; a second MEA comprising: a second anode positioned to receive the second gas stream from the first cathode of the first MEA, the second anode containing a catalyst adapted to oxidize at least some of second hydrogen gas content in the second gas stream to produce hydrogen ions and electrons; a second electrolyte positioned and adapted to receive and transfer at least some of the hydrogen ions produced by the second anode; and a second cathode positioned to receive at least some of the hydrogen ions transferred by the second electrolyte of the second MEA, the second cathode containing a catalyst adapted to reduce the at least some of the hydrogen ions to produce a third gas stream having a third hydrogen gas content greater than the first hydrogen gas content and a third impurity gas content less than the first impurity gas content; and at least one passage between the first electrolyte and the second electrolyte for introducing hydrogen-containing gas to the second gas stream, the at least one passage comprising voids between mating surfaces of the first cathode and the second anode.

21. The hydrogen gas purifier cell as recited in claim 20, wherein the purifier cell further comprises at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second gas stream.

22. The hydrogen gas purifier cell as recited in claim 21, wherein the at least one passage between the first electrolyte and the second electrolyte for discharging at least some of the second gas stream comprises the voids between the mating surfaces of the first cathode and the second anode.
